

# 256-row tSZ diagnostic

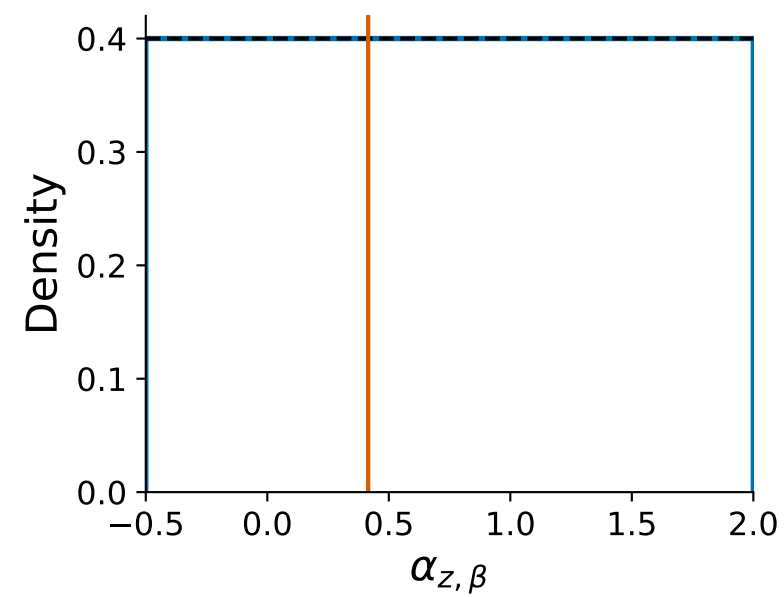
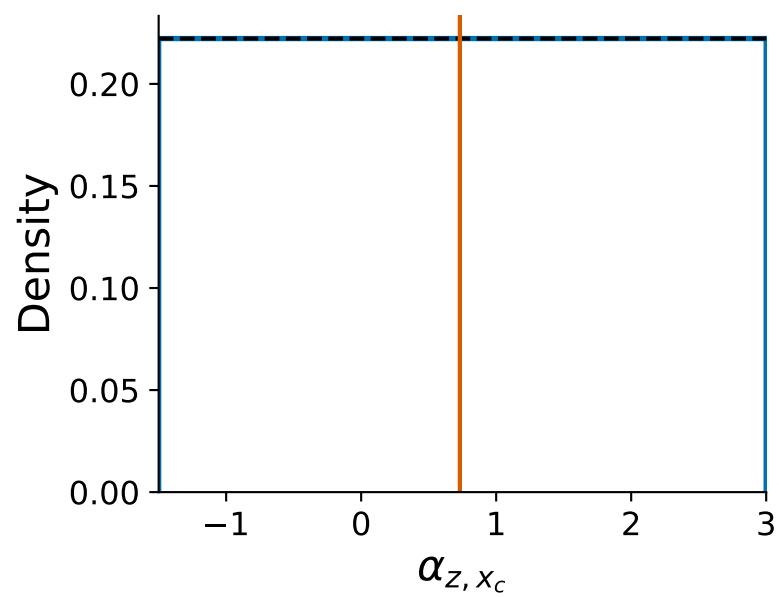
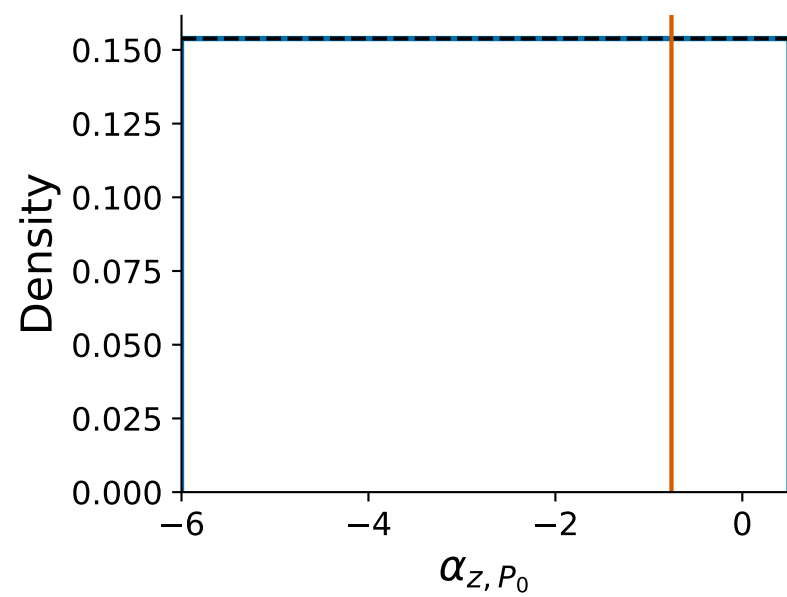
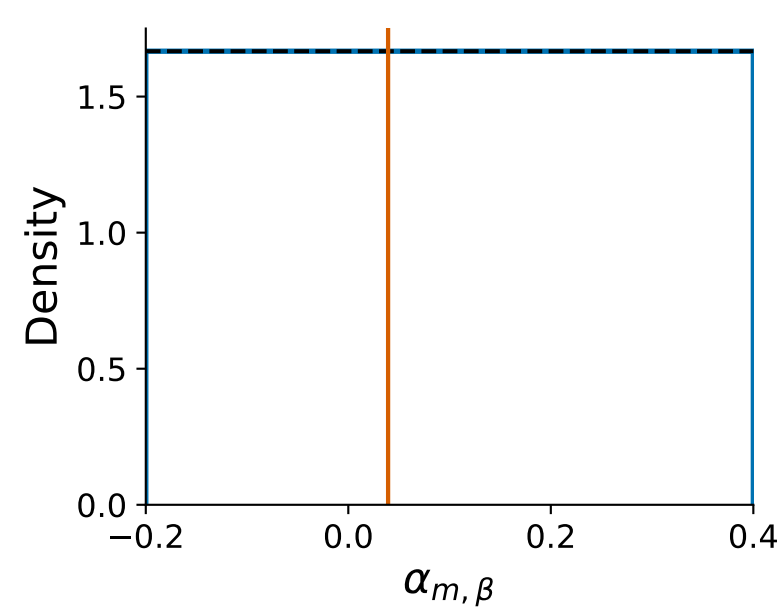
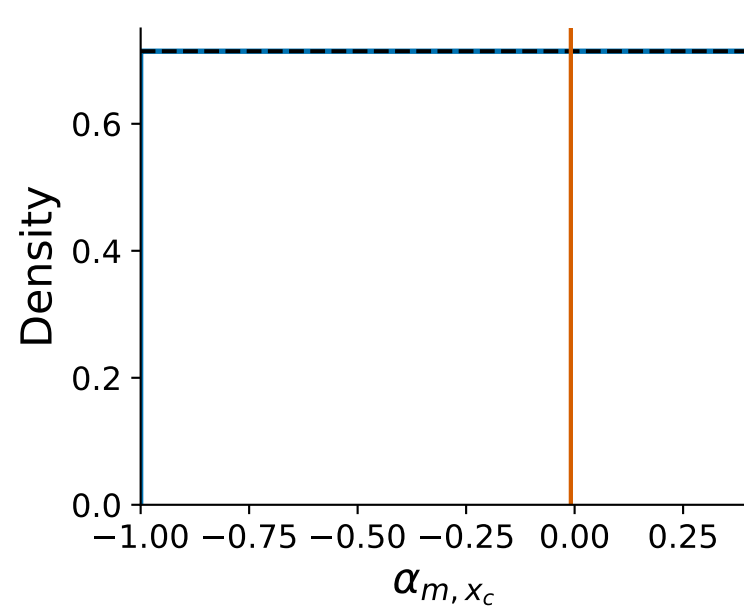
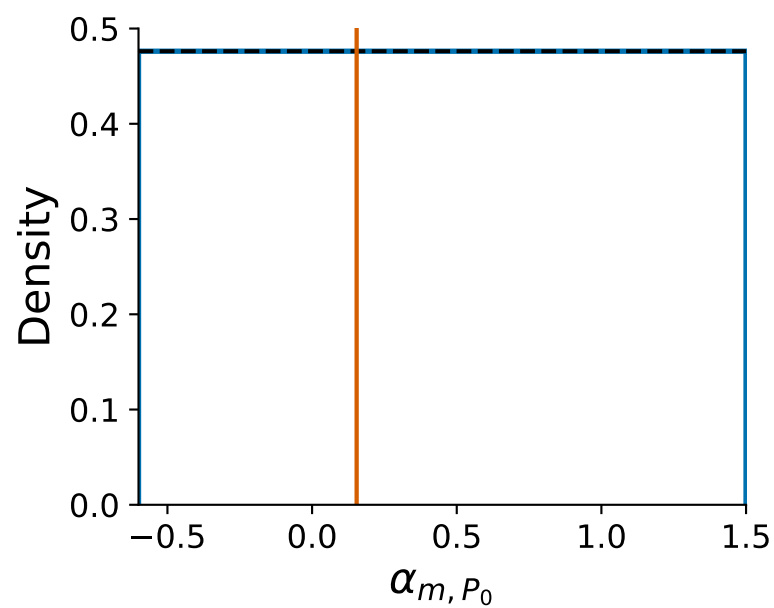
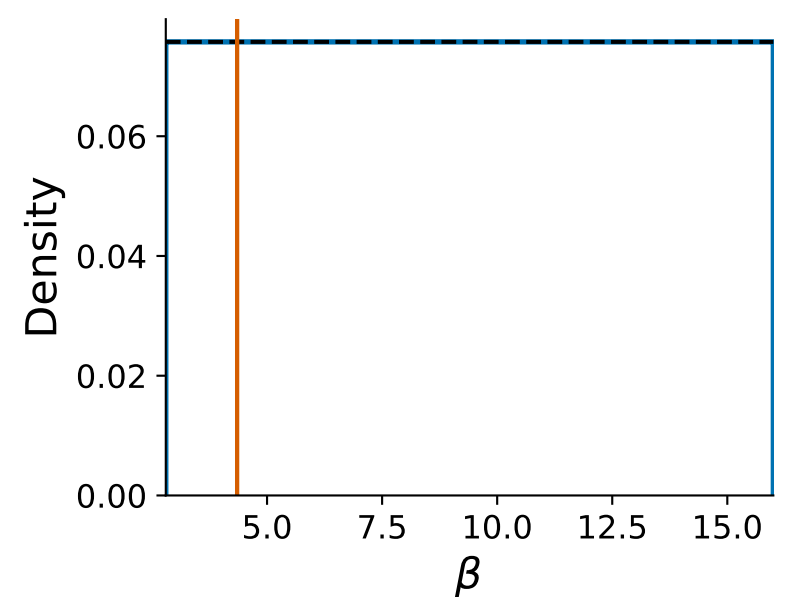
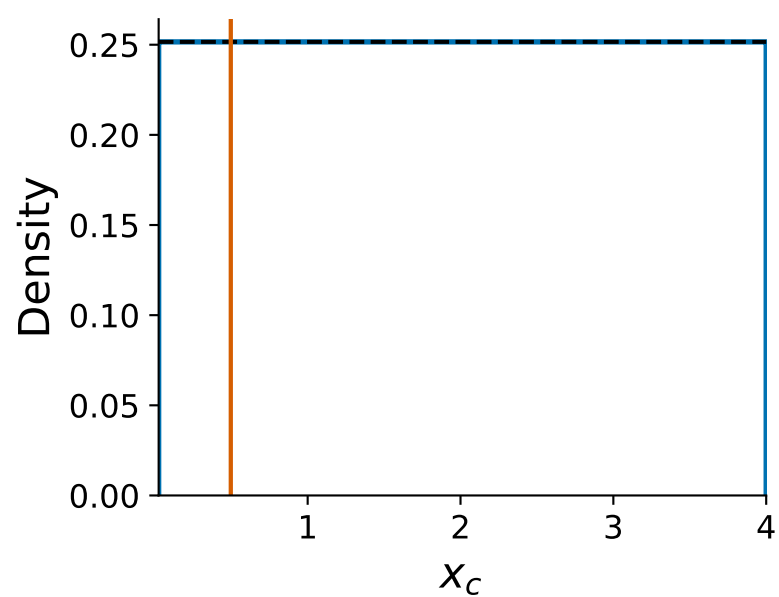
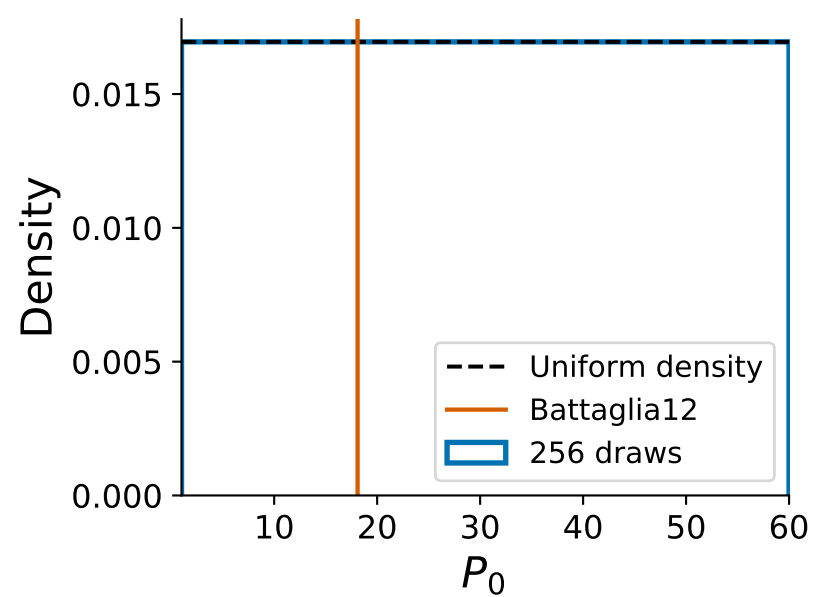
Completed 256 saved parameter draws, with no prior rejection. The primary inference uses all 7,900 individual multipoles,  $\ell = 80 \dots 7979$ , before local nine-direction MOPED compression. The split is 192 training-pool rows and 64 held-out rows.

Physics: spherical gas support at 4 R200c, exact finite line-of-sight chord, fixed painter cosmology  $h=0.68$ ,  $\Omega_b=0.049$ ,  $\Omega_c=0.261$ , and the selected 85,224,251-halo catalogue. A 2 arcmin Gaussian smooths the signal once. The  $f_{\text{sky}}=0.4$  cap, 60 arcmin apodization and mask seed 12345 are shared. Each row has two independent SO noise splits.

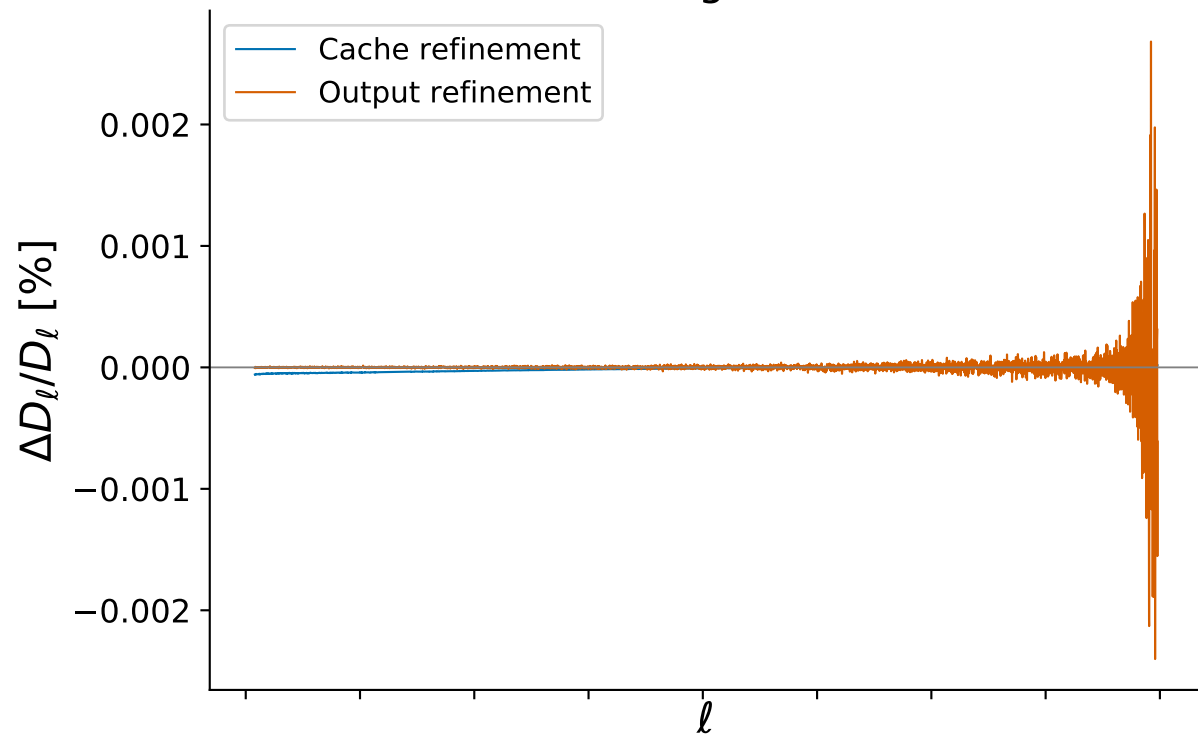
The accelerated-versus-reference numerical gate passed: True. Cache choices: {'256x128x64': 215, '512x256x128': 41}. Every row was checked at 2,496 direct spherical-column points; four selected prior rows also received a finer-cache full-sky comparison. The clean per-multipole target is 1%. These sampled tests do not prove a uniform bound over all continuous halo coordinates.

There were 0 incomplete held-out posterior-sampling runs. The following pages show the prior baseline, shuffled-data baseline, learning curves and approximate marginal coverage. No larger dataset is submitted by this pipeline.

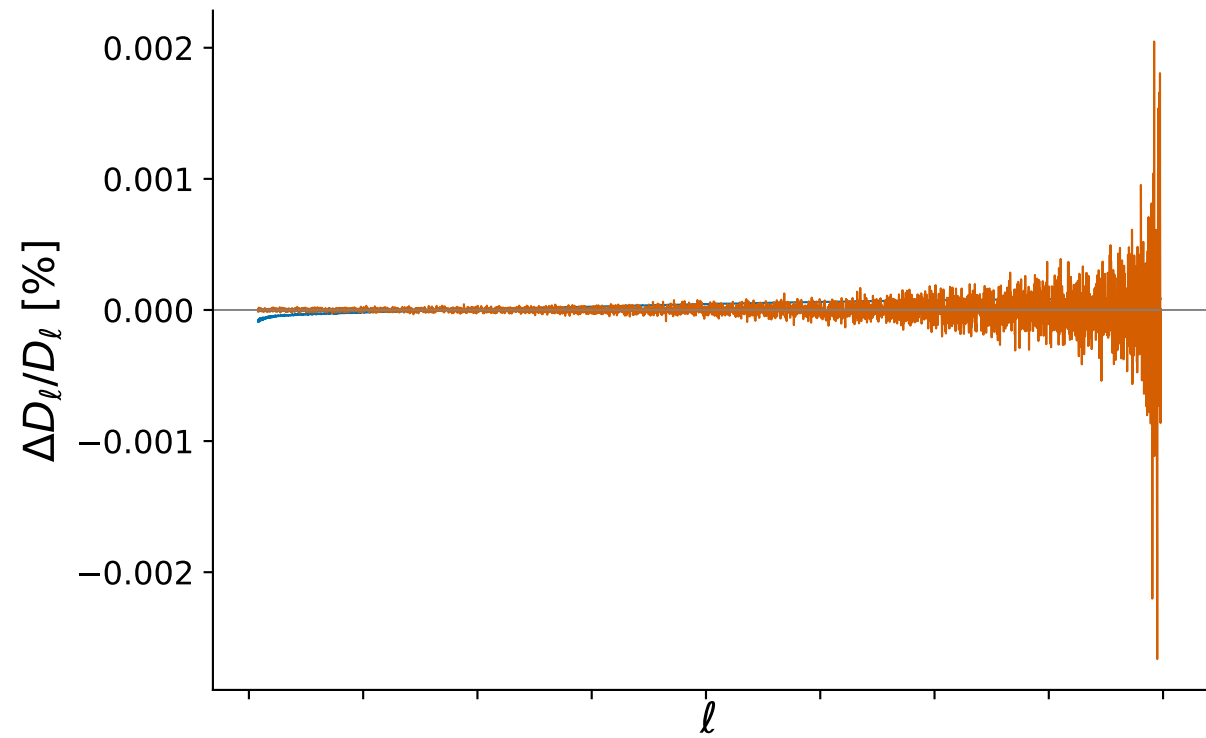
A percentage accuracy target and an inference error are different. Existing 8192-versus-16384 references give B12 a 0.091% spectrum-norm residual, a 1.06% worst-multipole residual and about 0.39 conditional-noise units in local nine-direction MOPED. The deliberately bright/shallow reference has a 0.0042% norm residual but about 2.33 conditional-noise units. Those discretization errors remain even when cache errors are negligible.



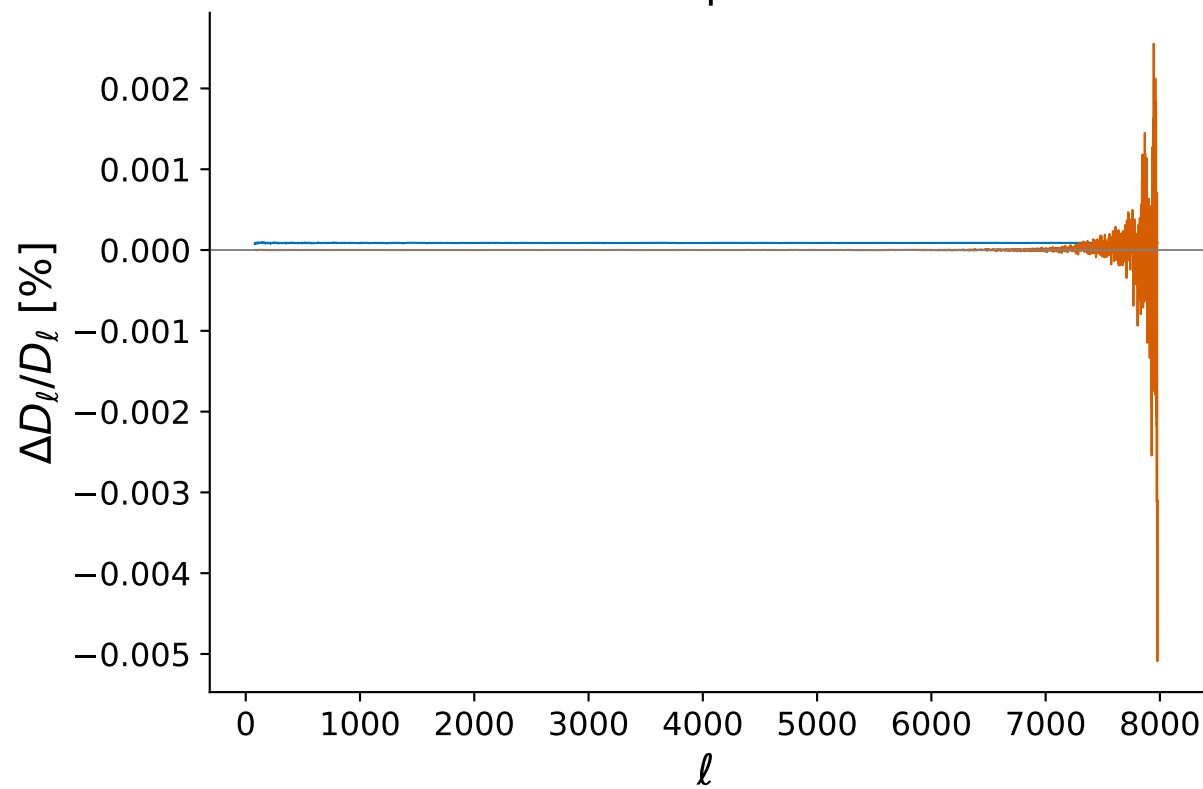
Battaglia12



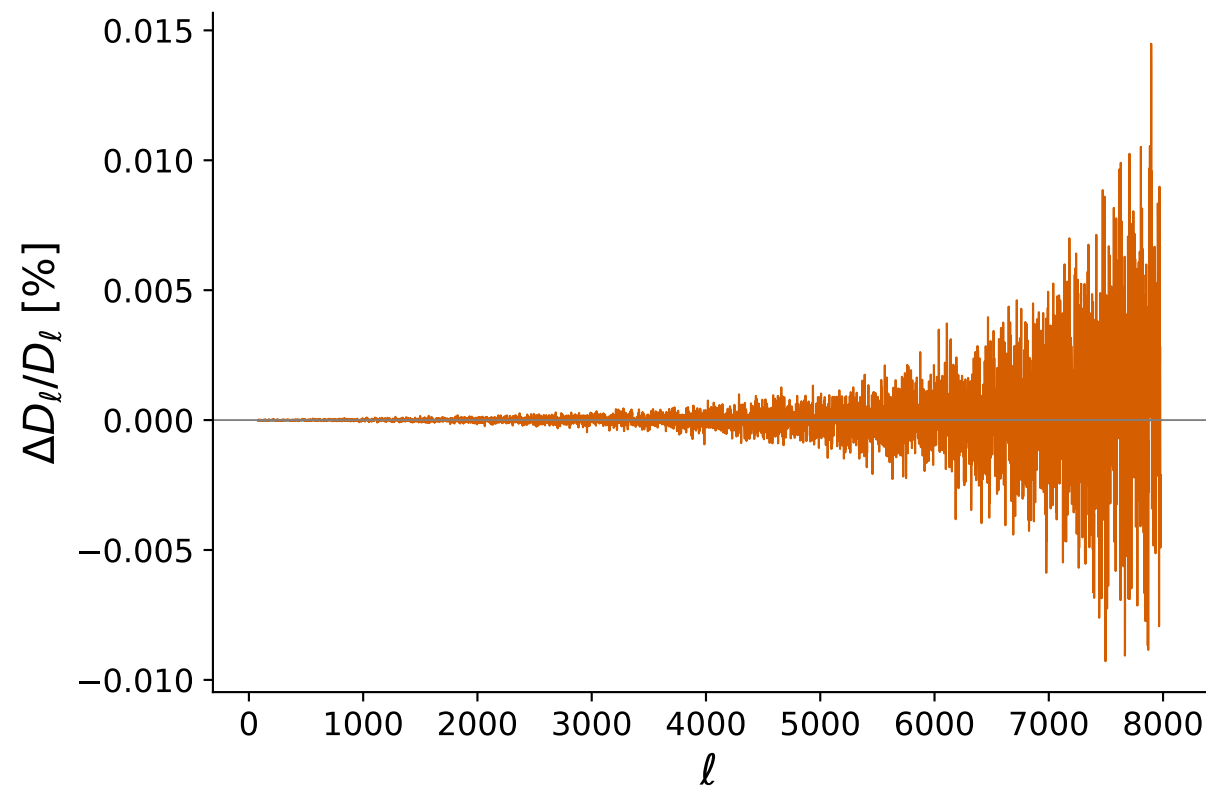
FL L1 m9

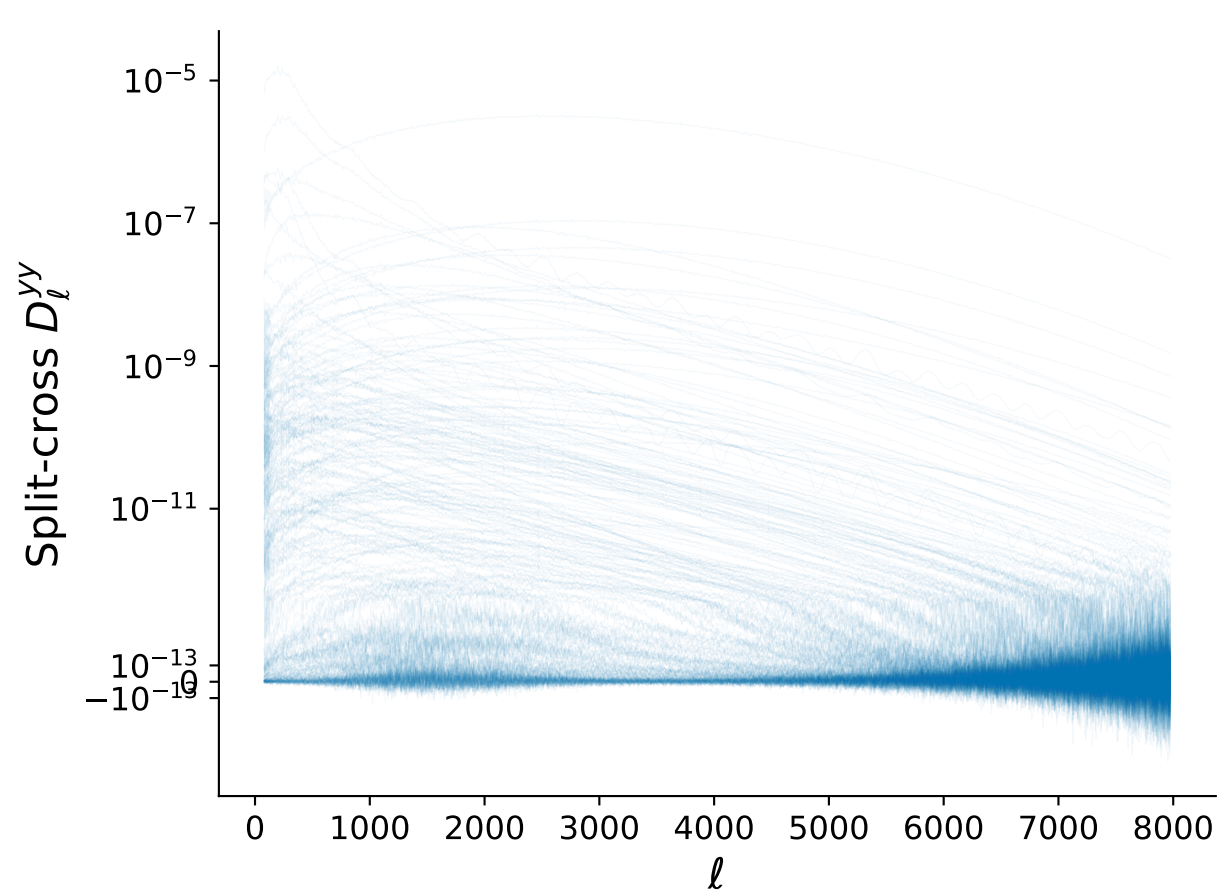
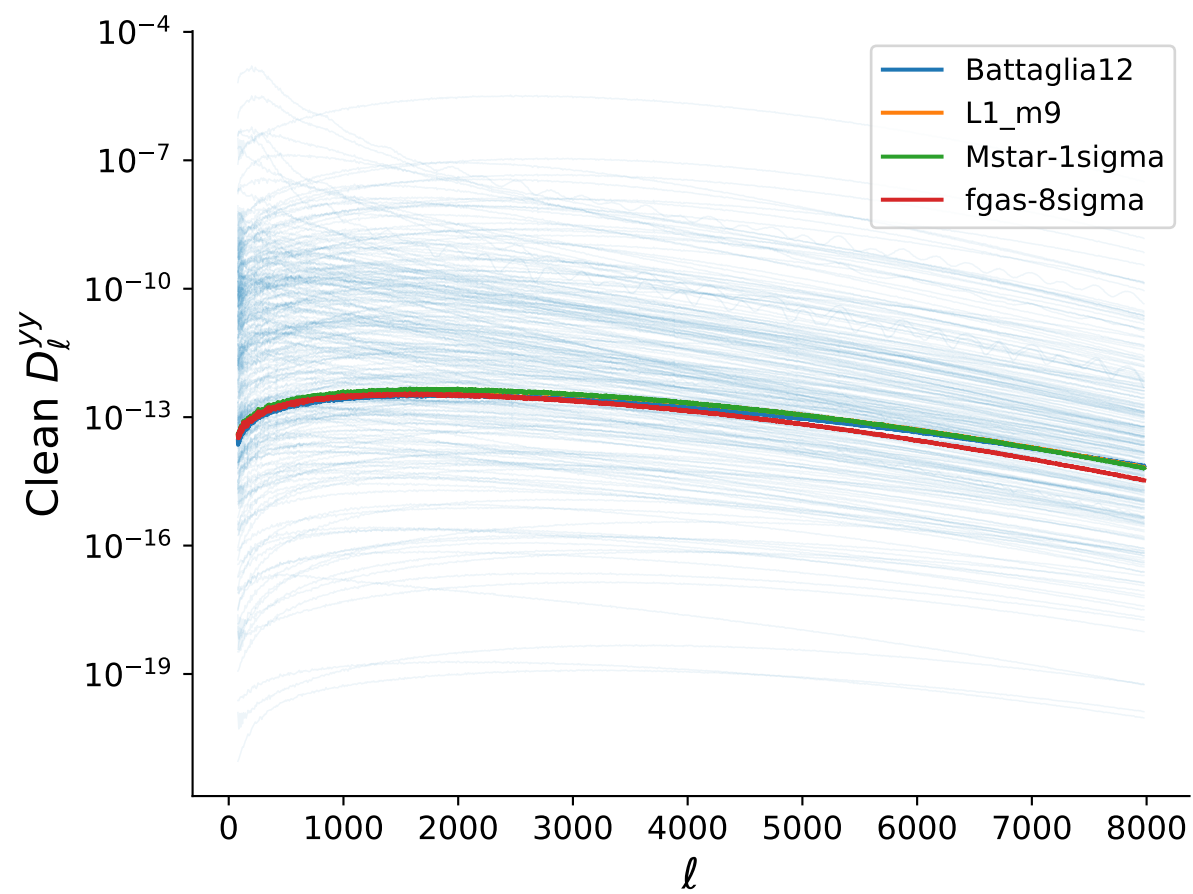


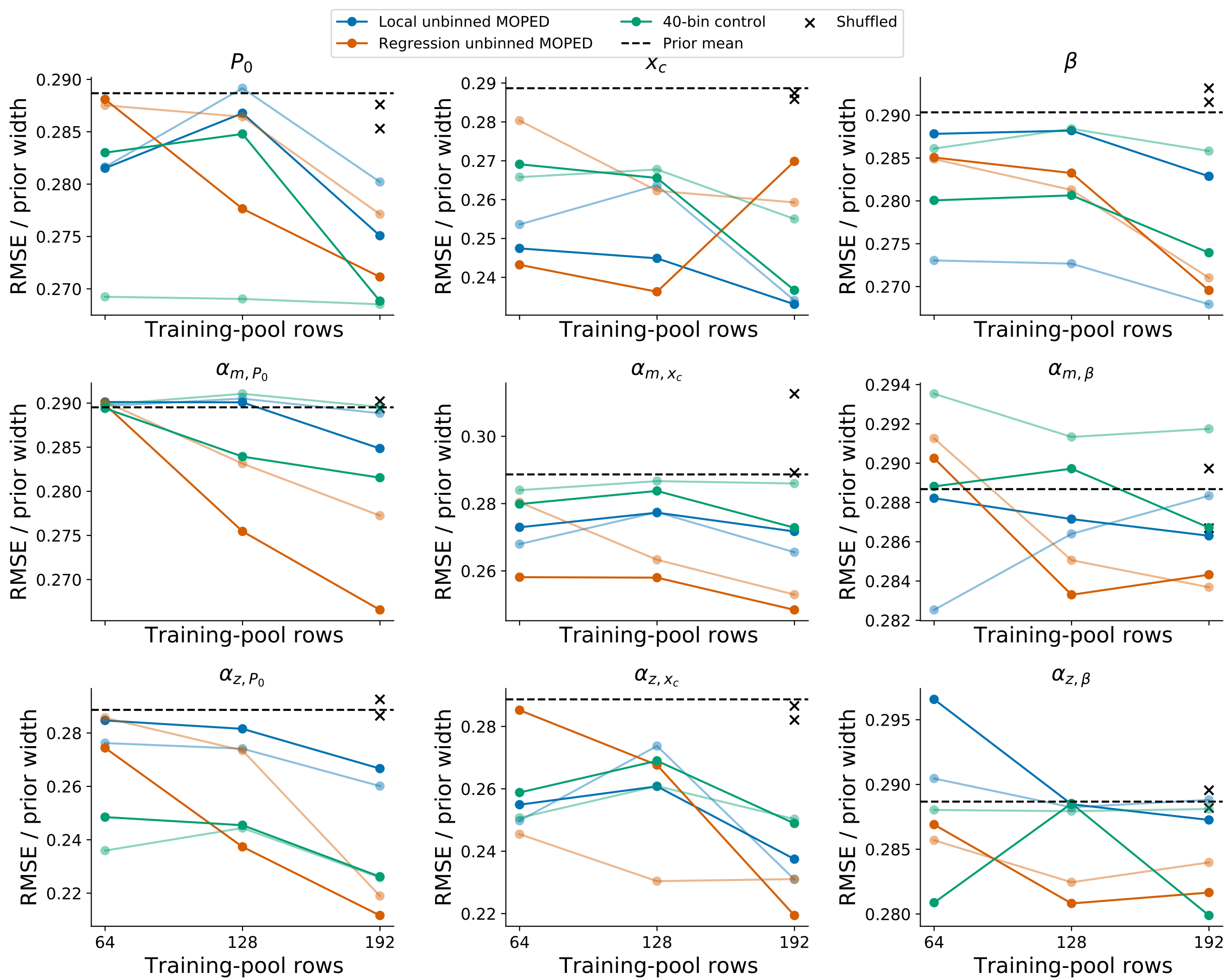
compact



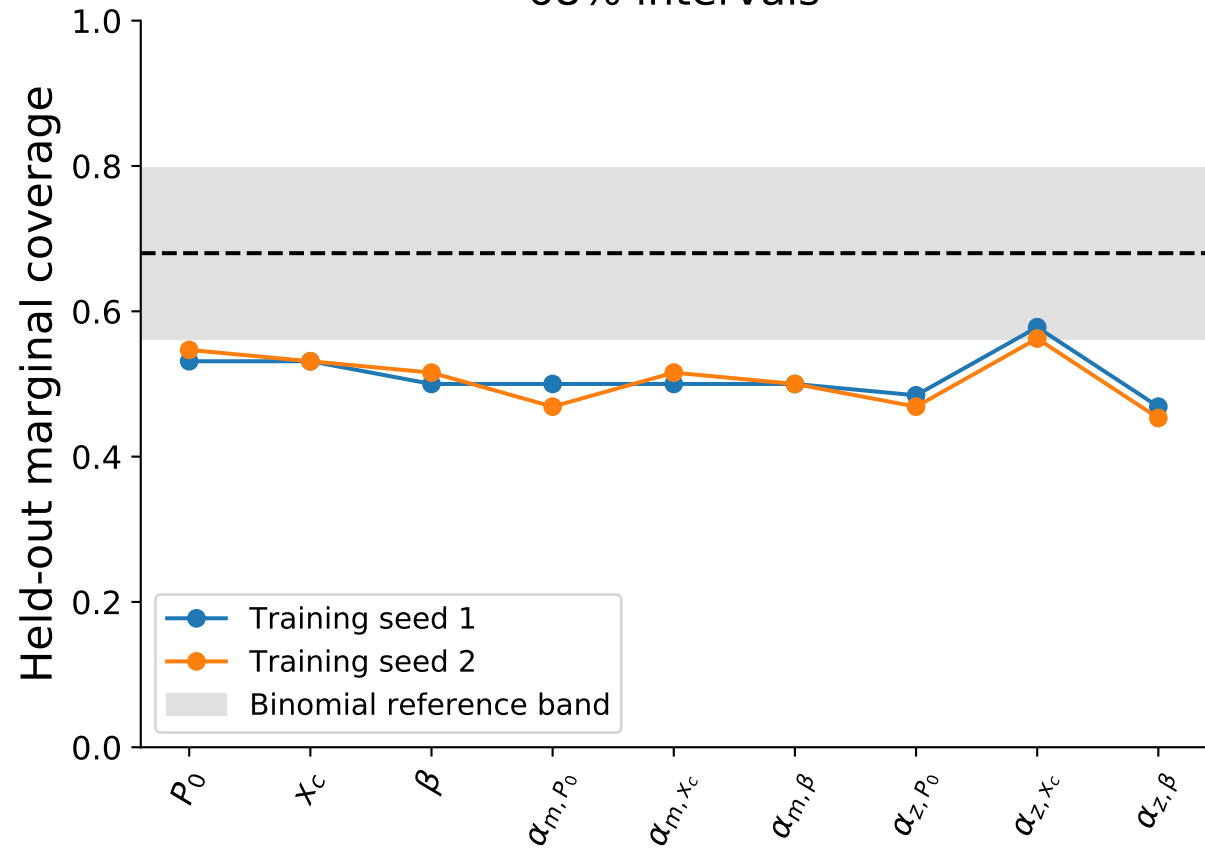
extended shallow



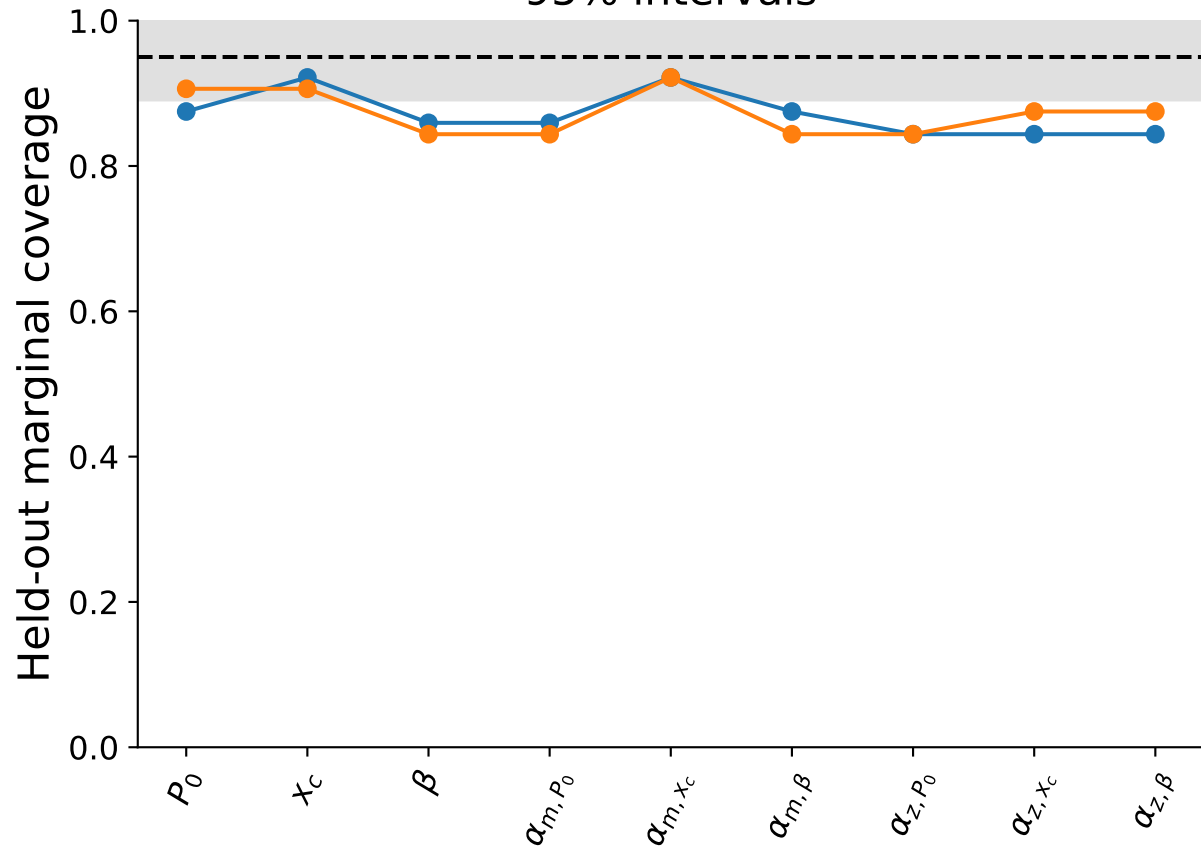




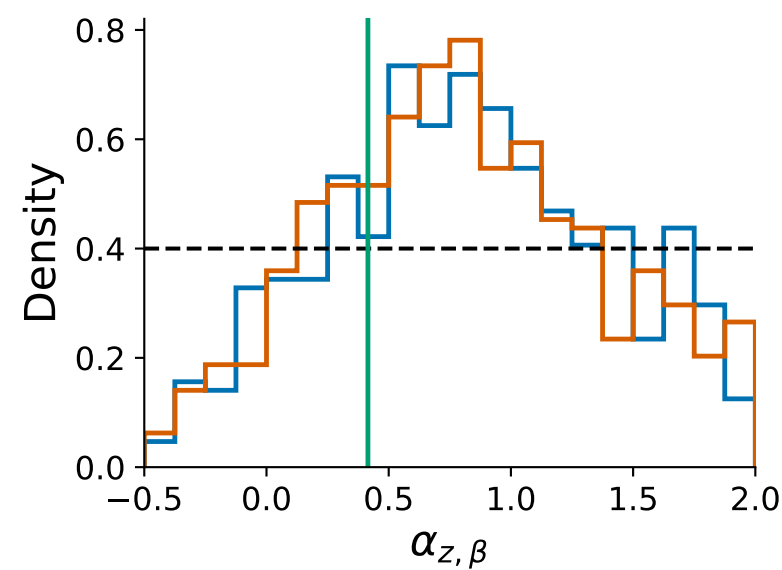
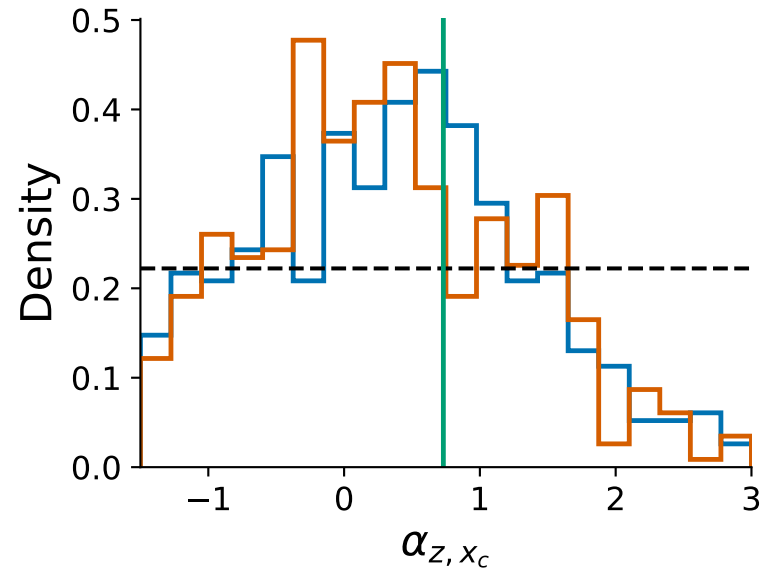
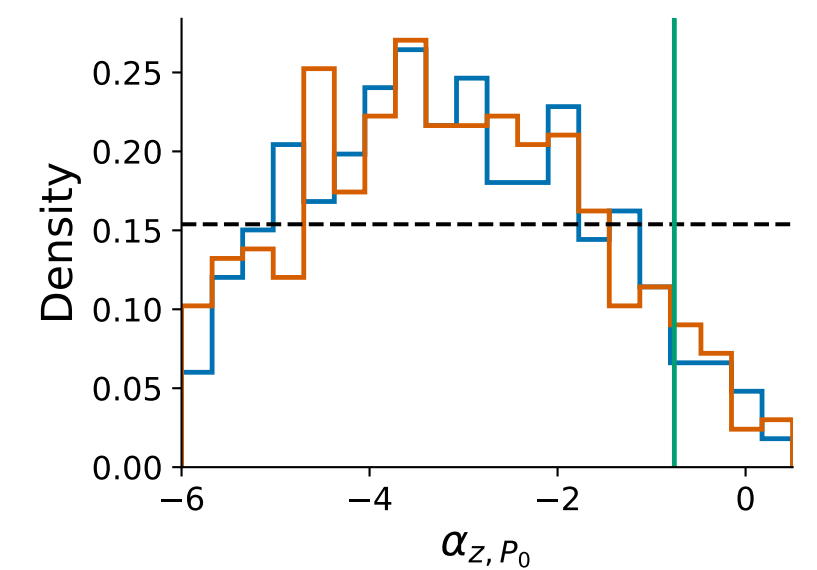
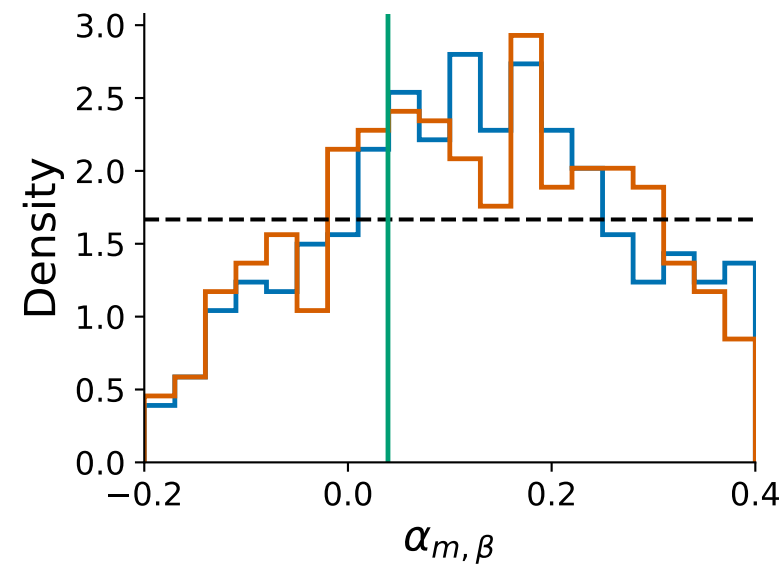
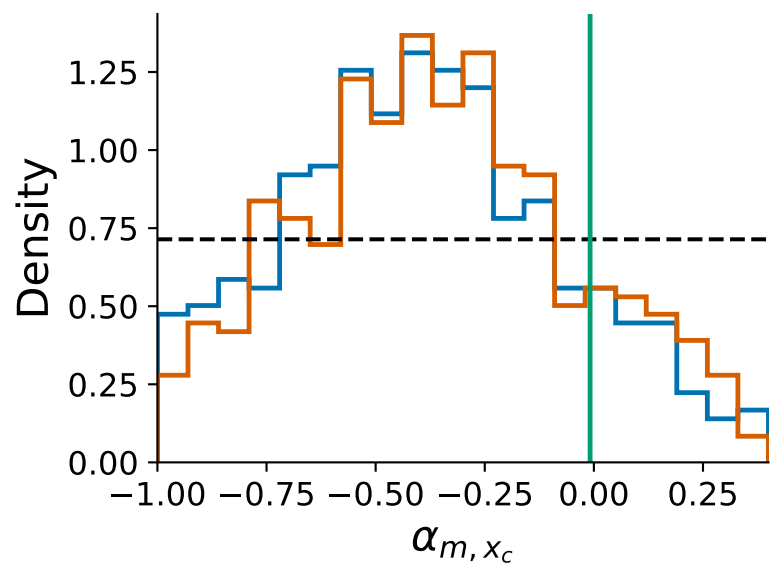
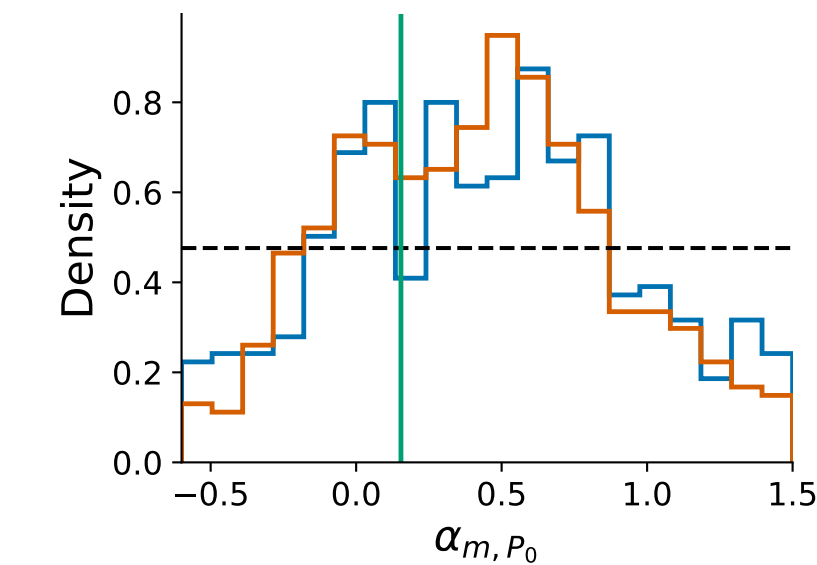
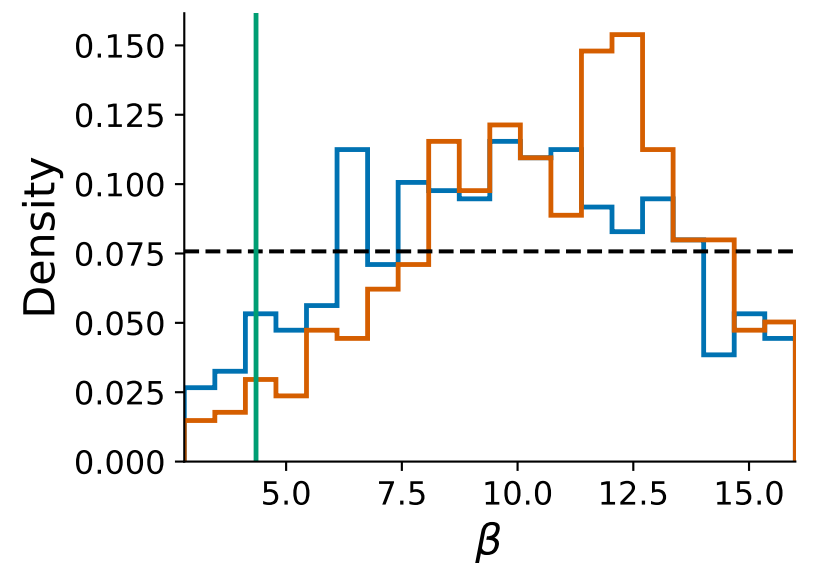
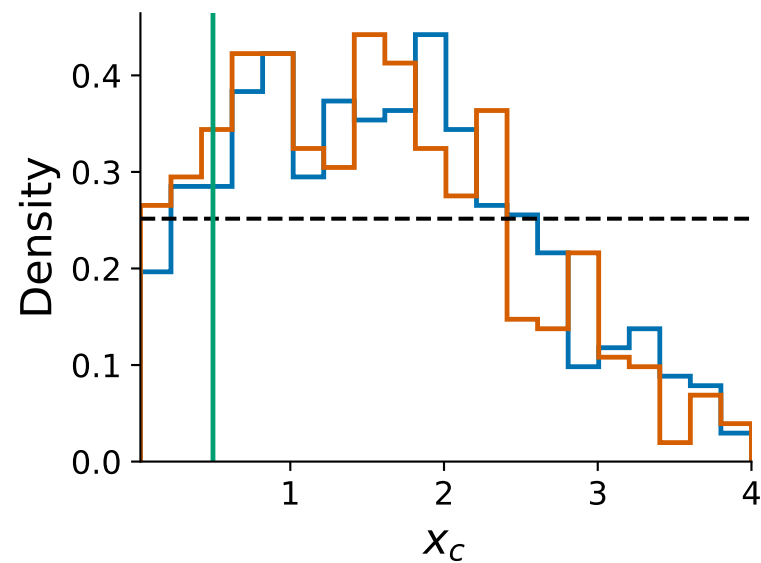
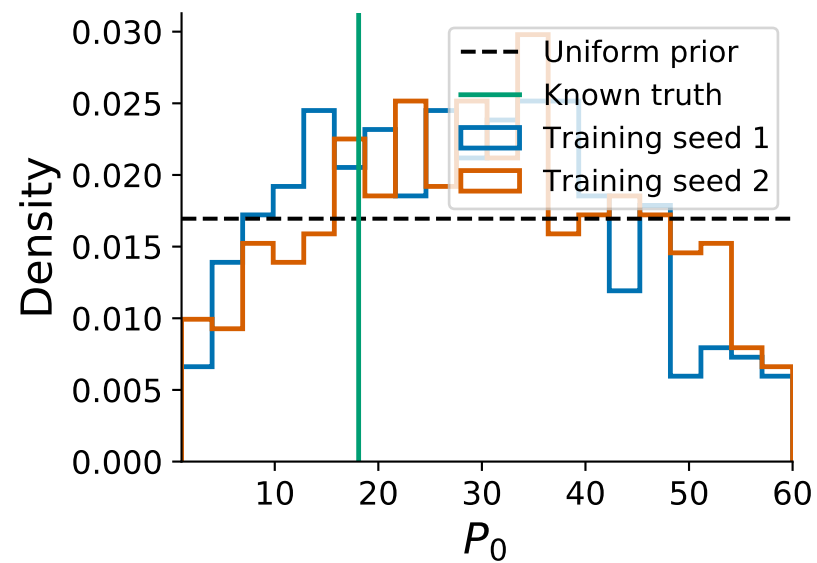
68% intervals



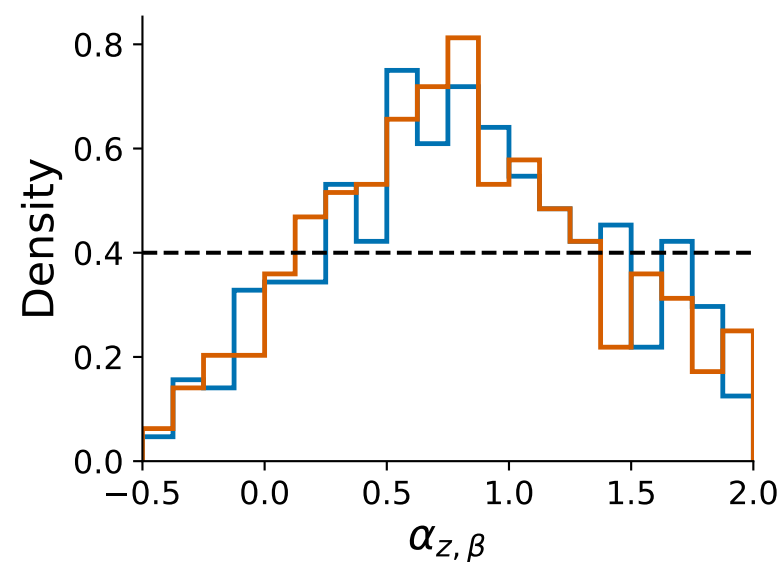
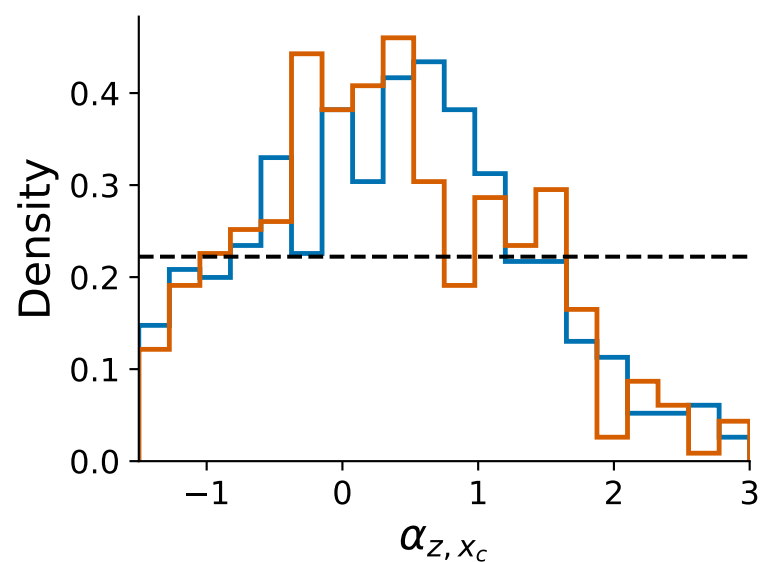
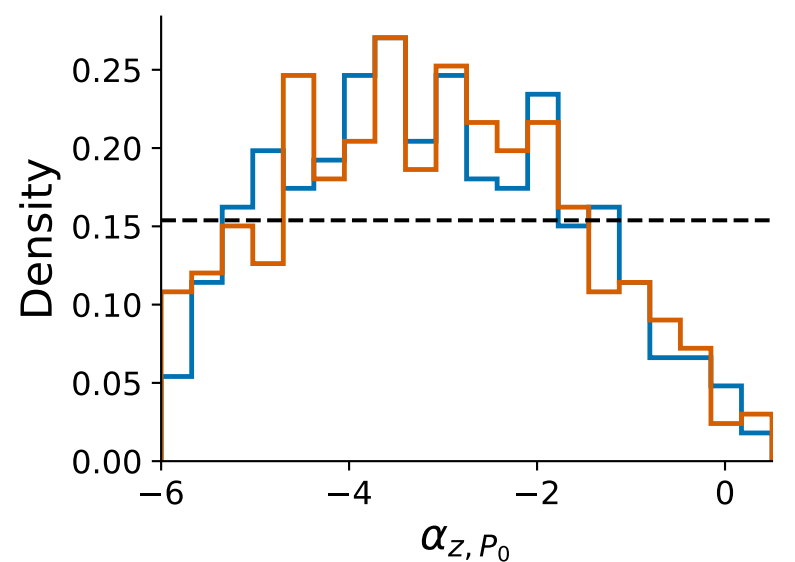
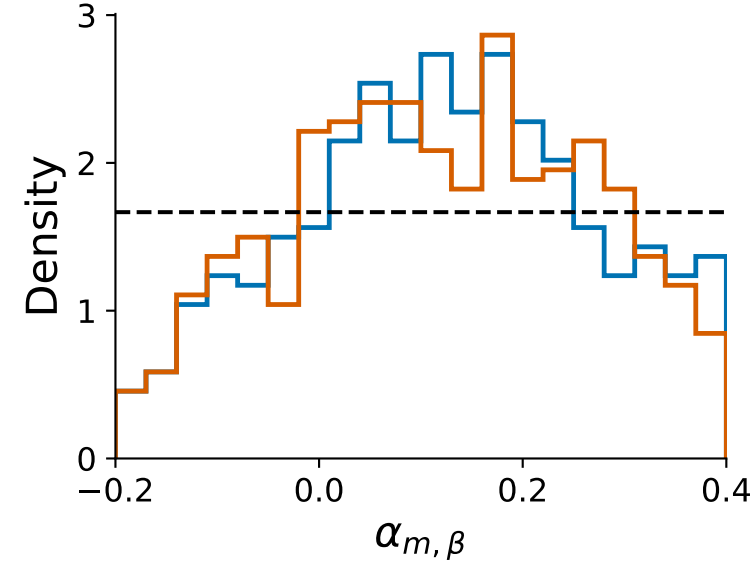
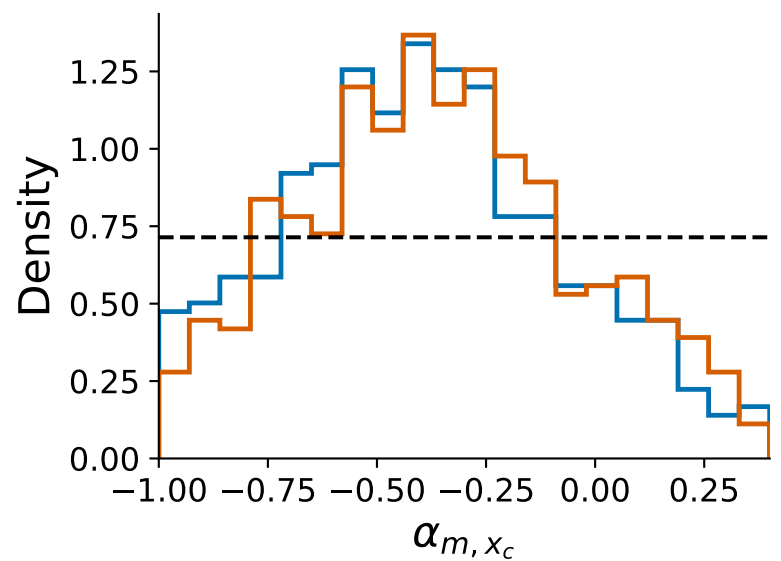
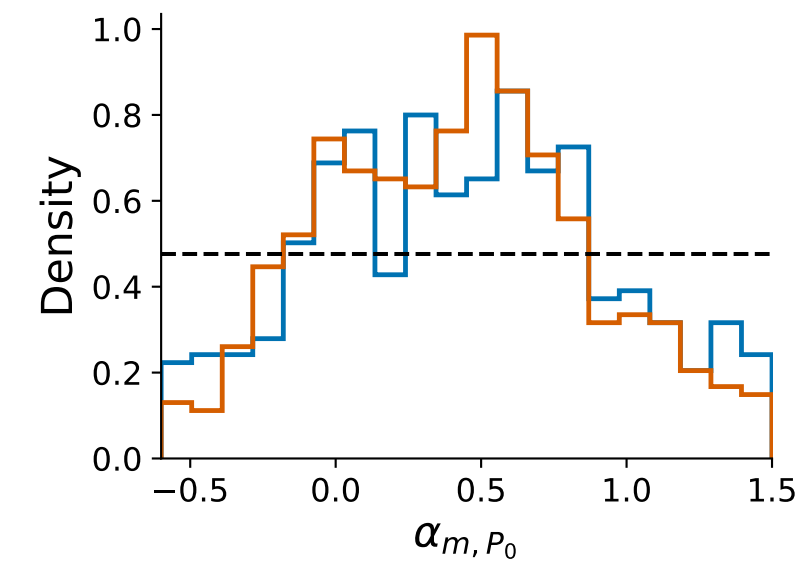
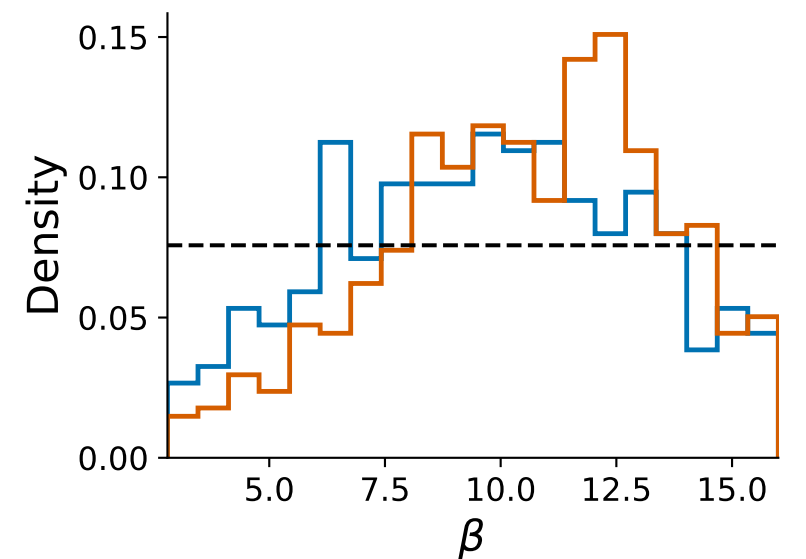
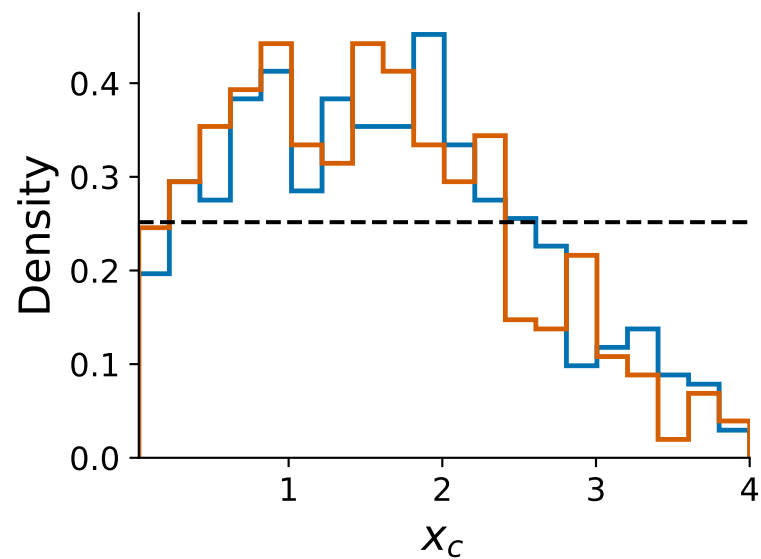
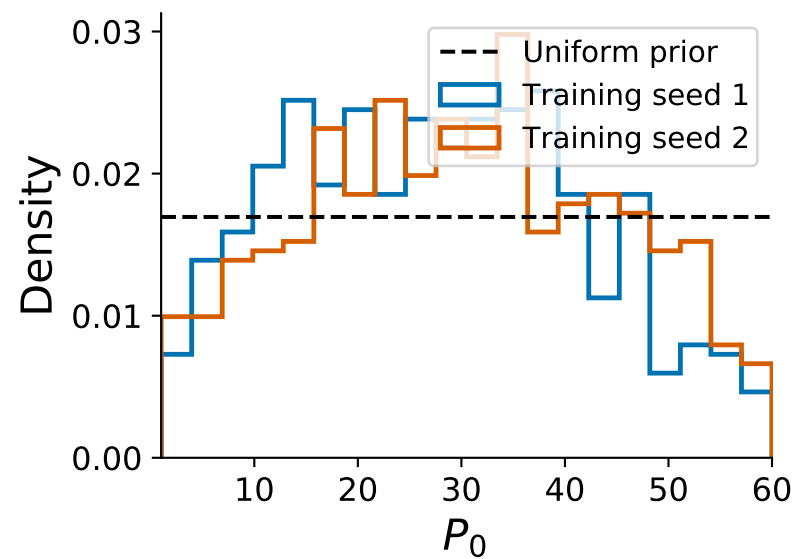
95% intervals



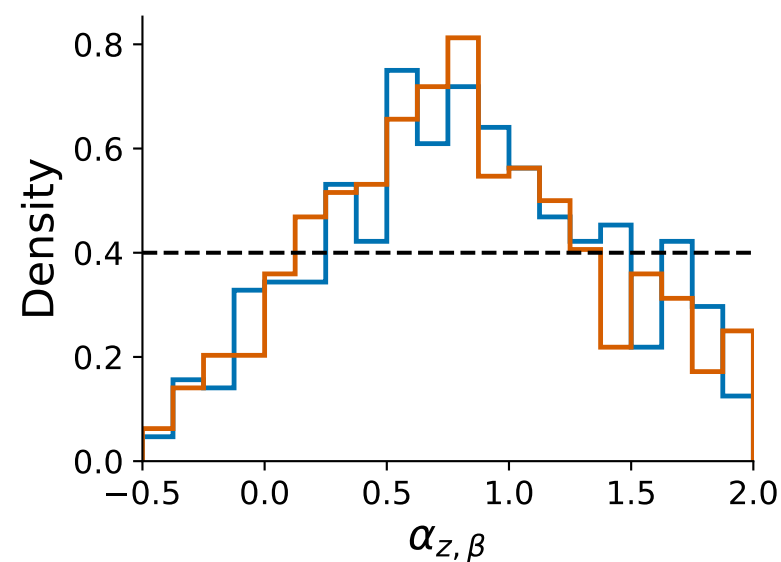
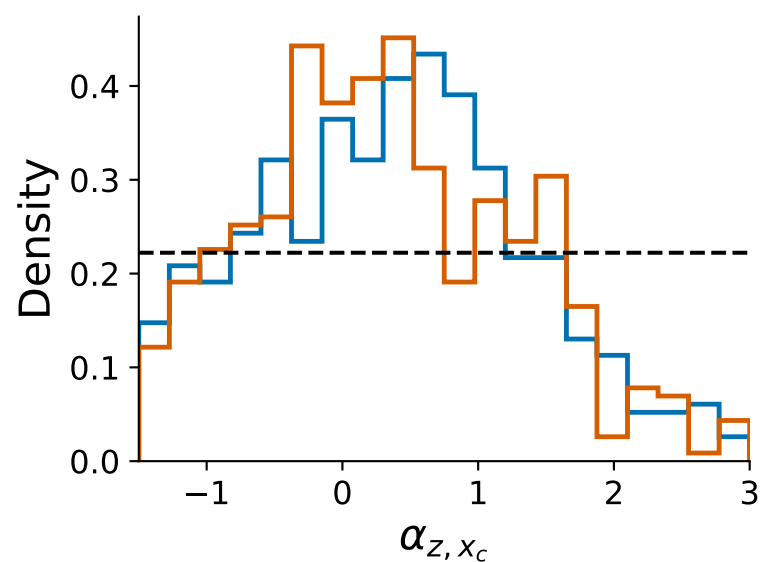
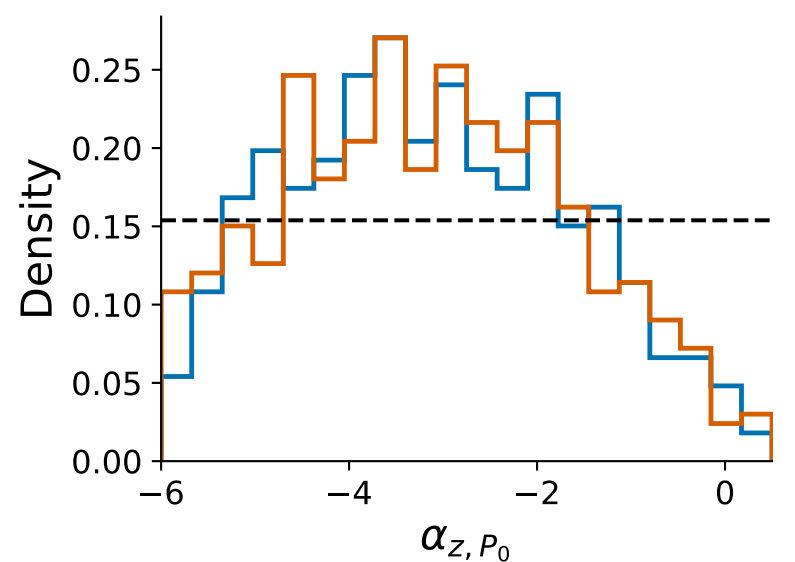
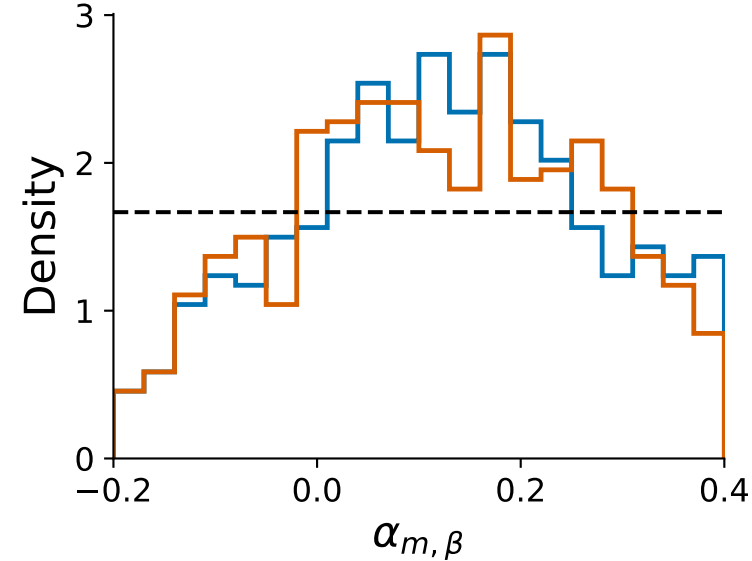
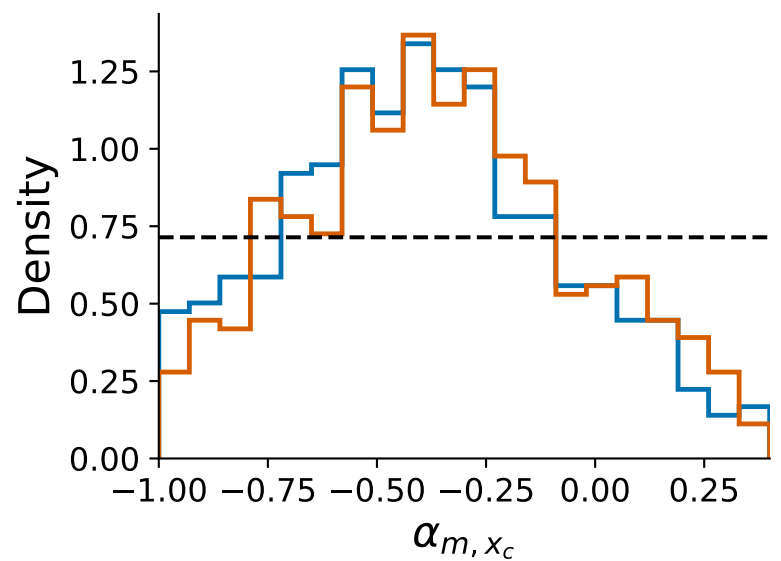
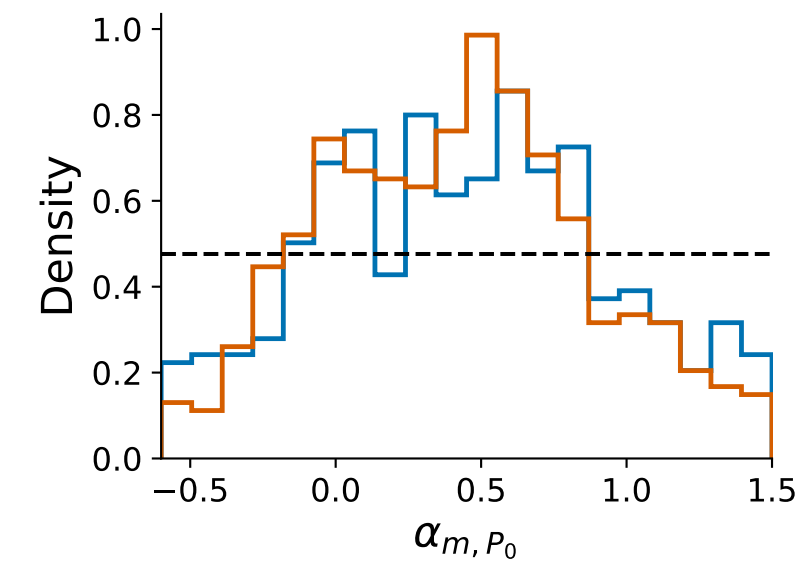
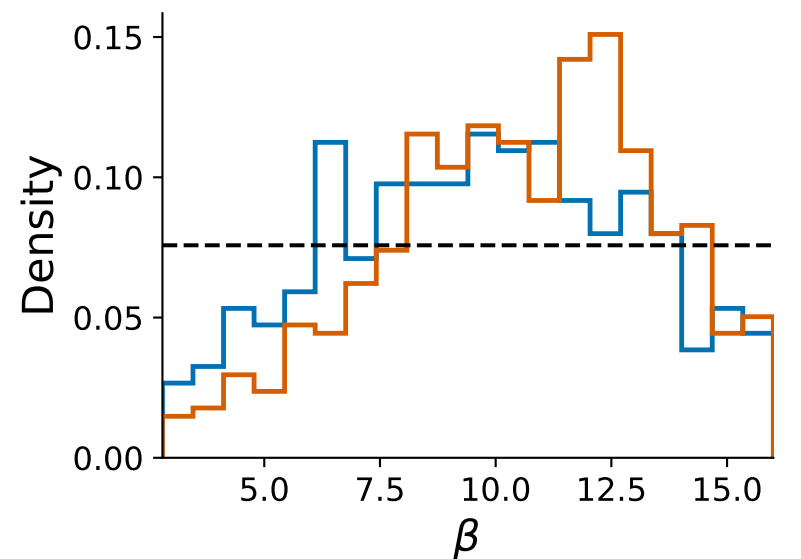
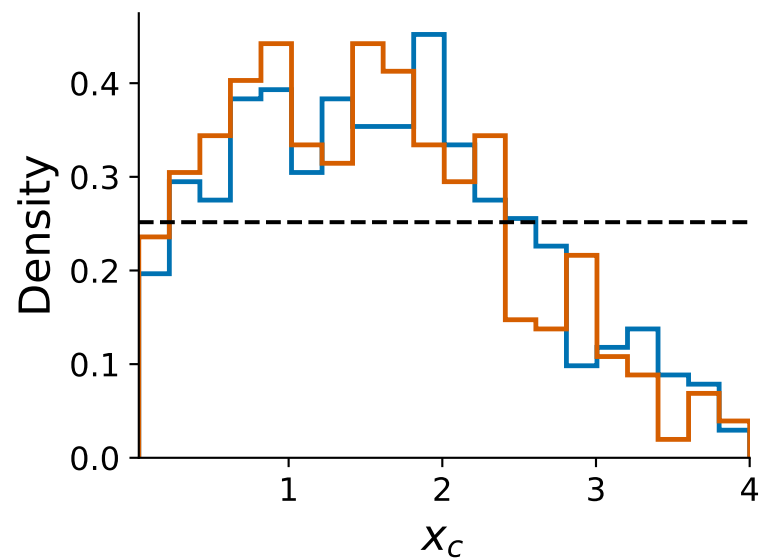
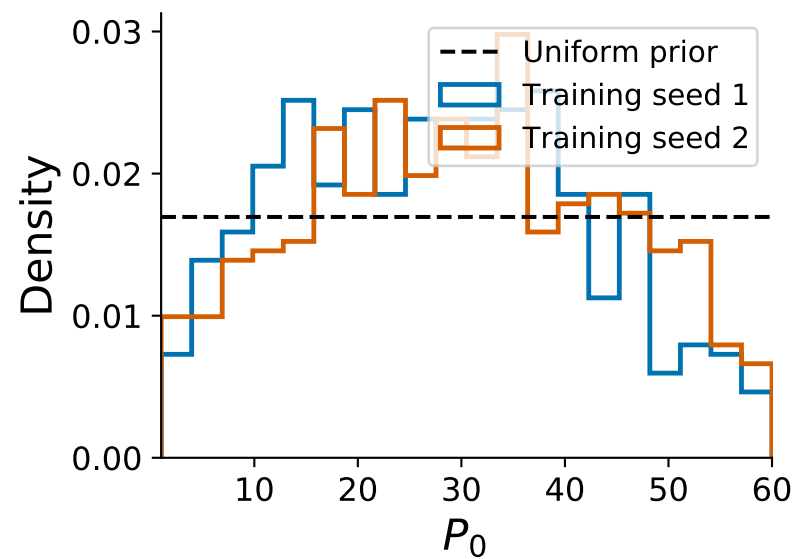
# Battaglia12



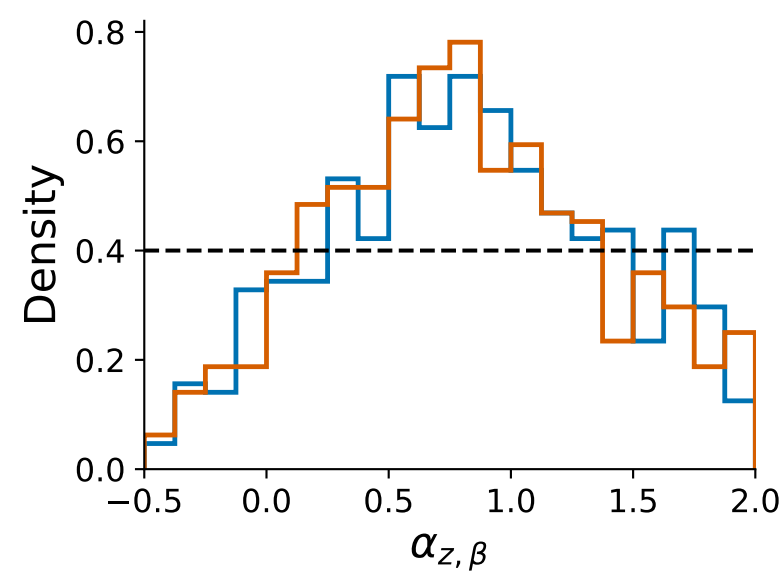
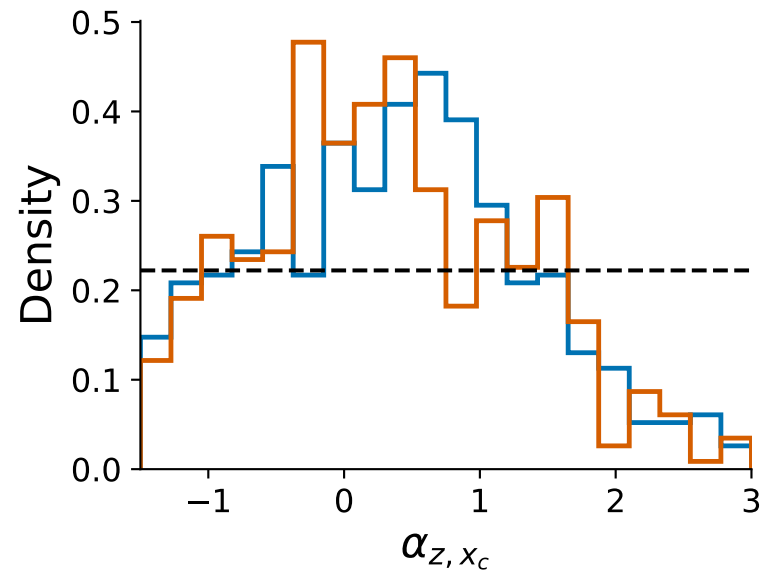
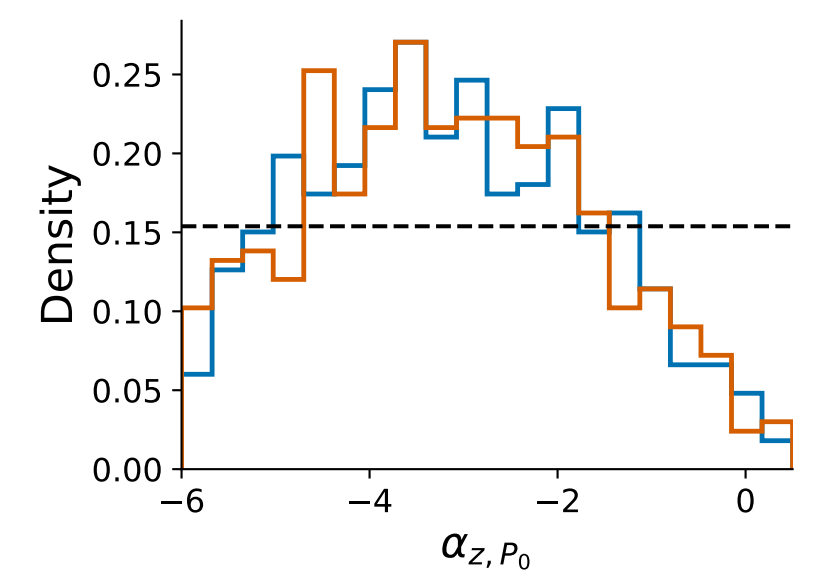
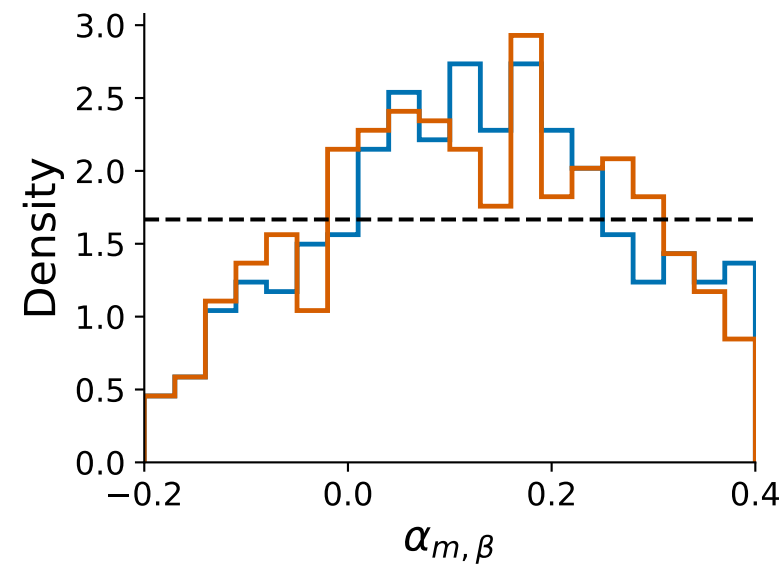
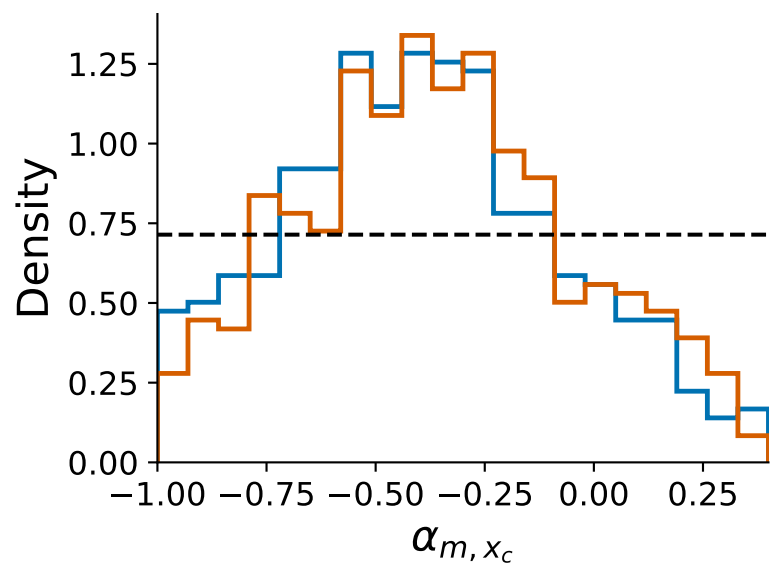
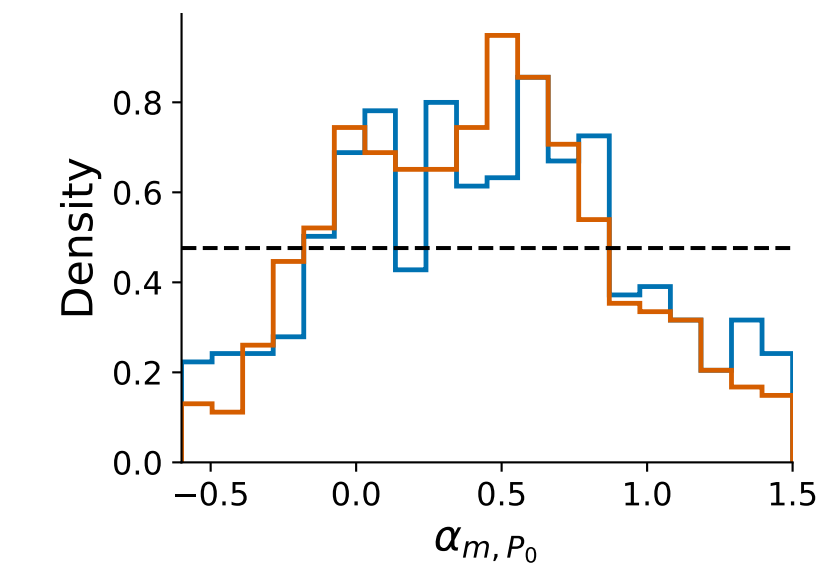
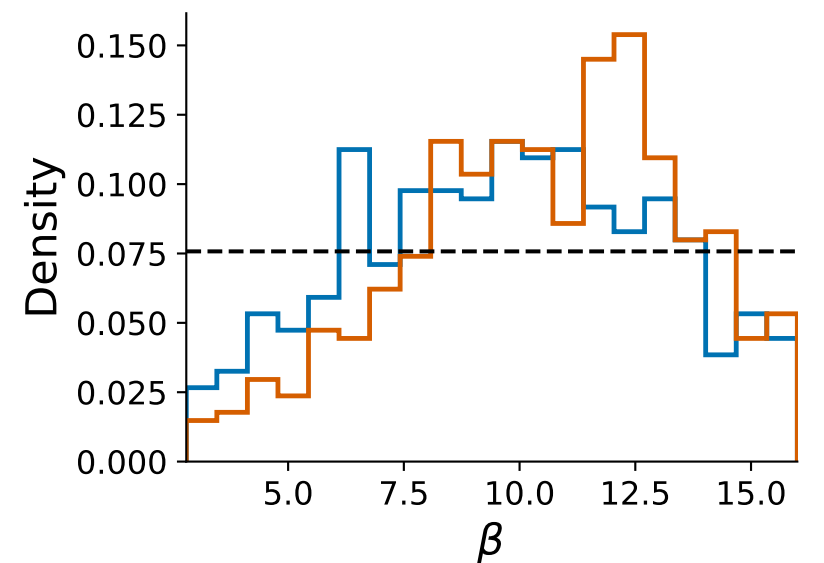
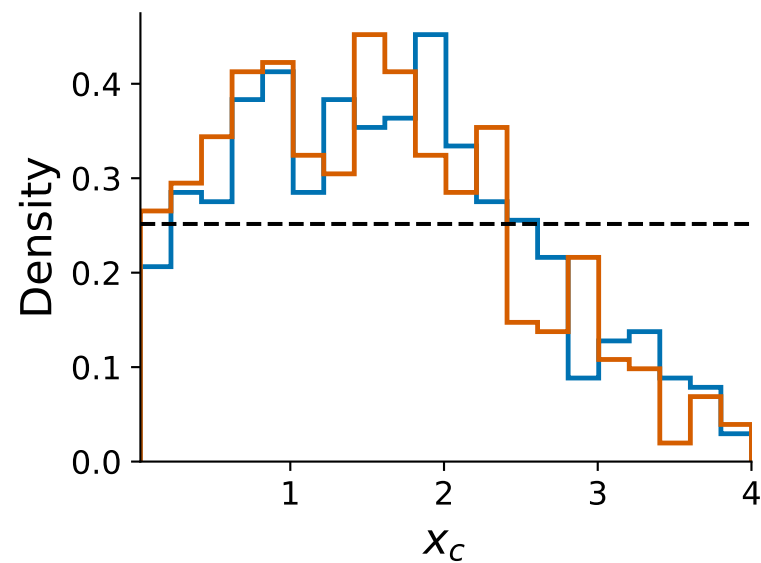
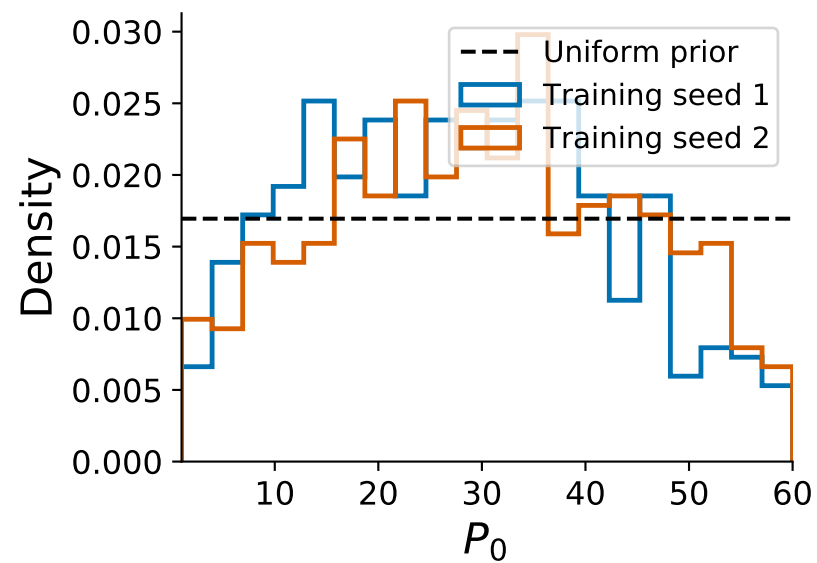
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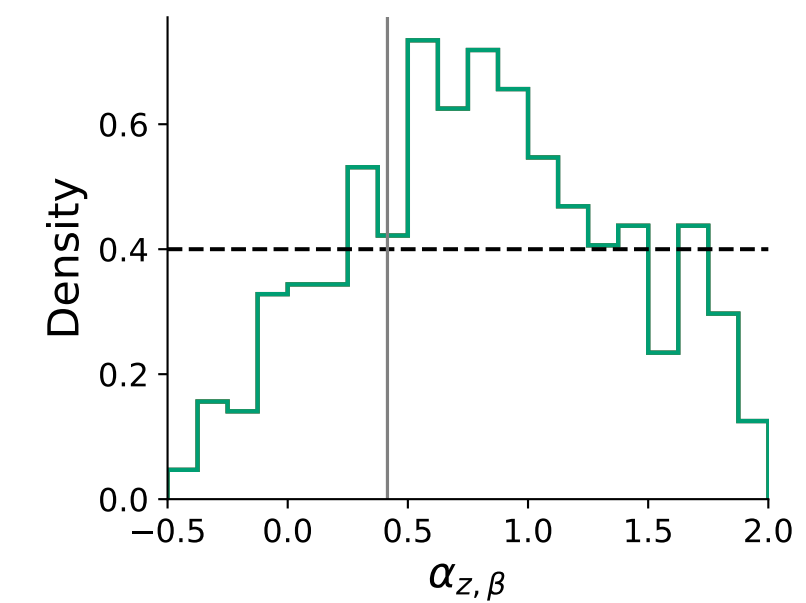
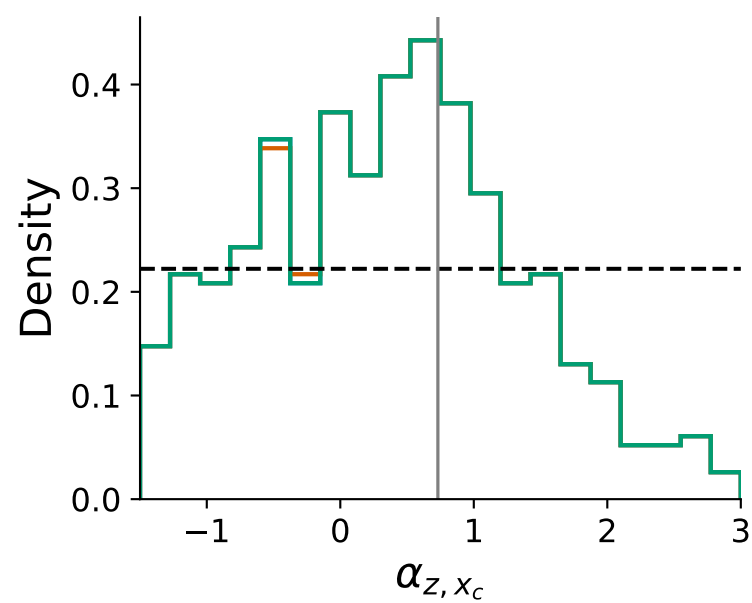
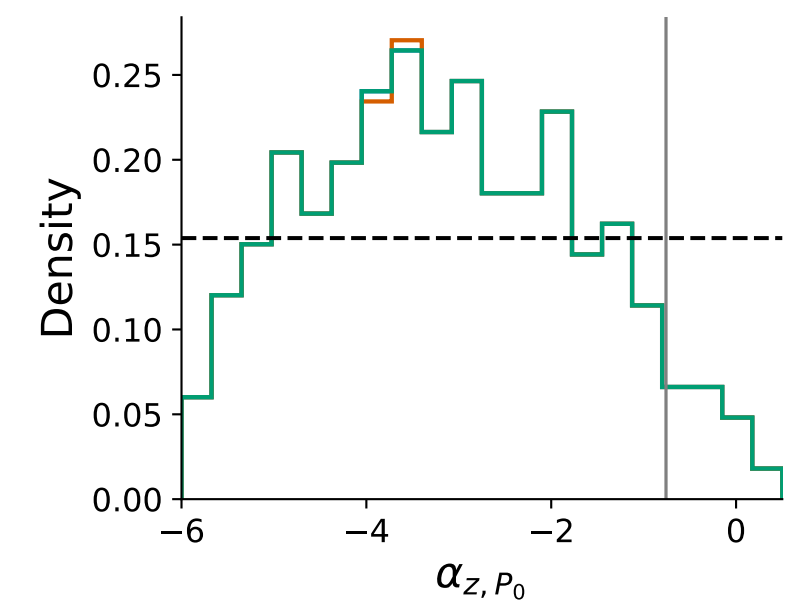
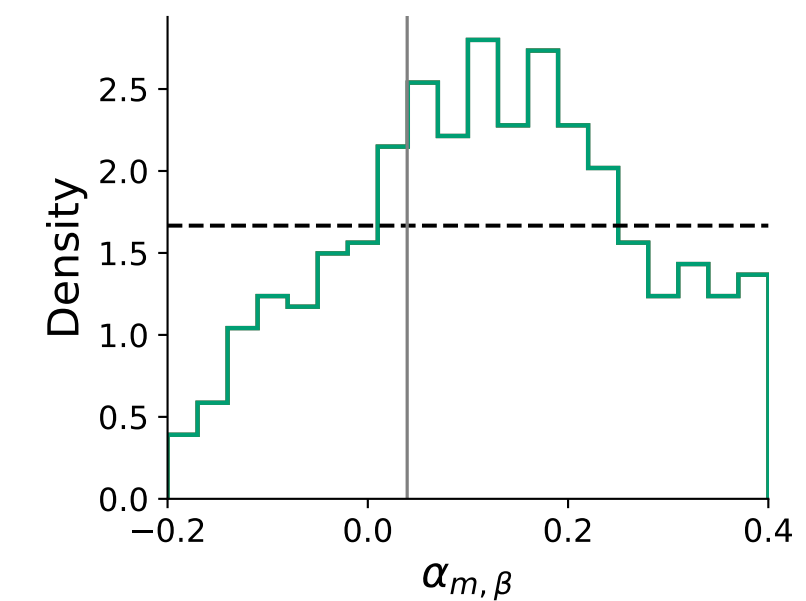
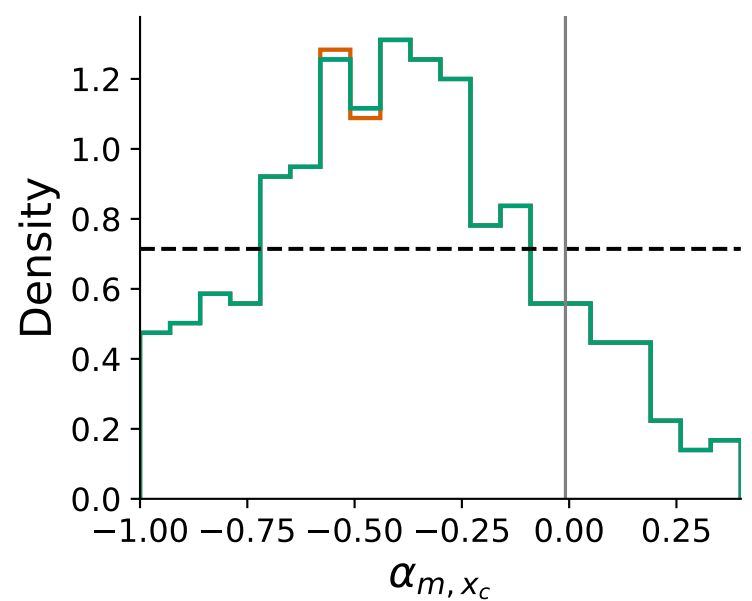
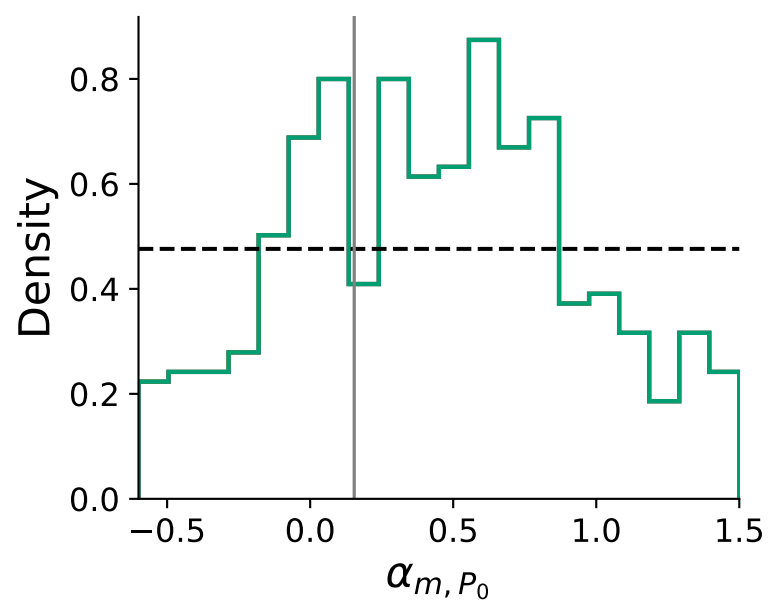
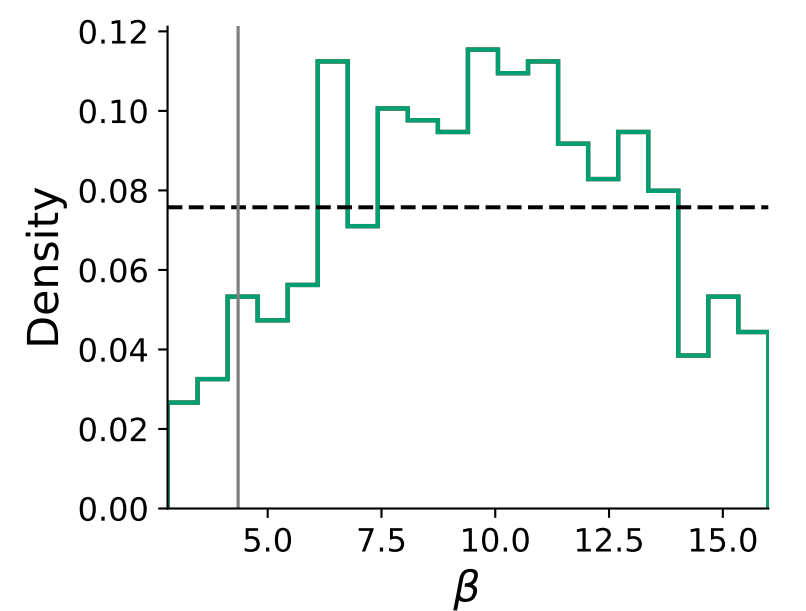
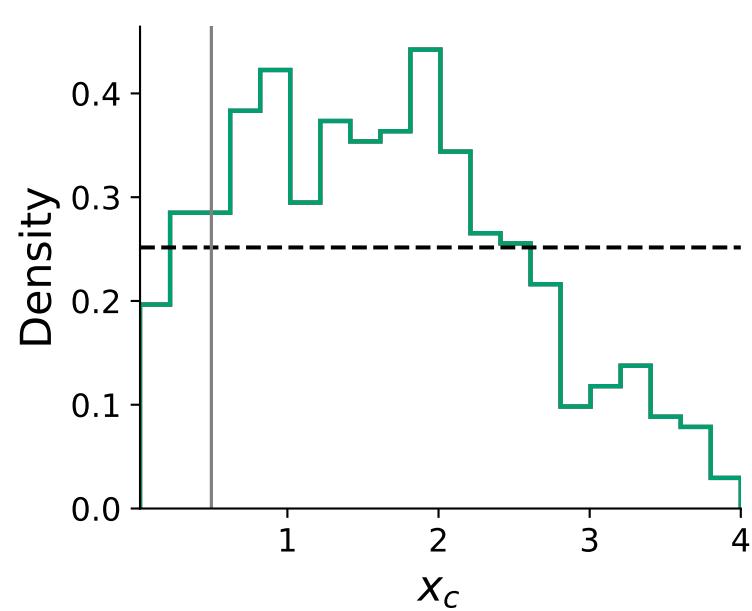
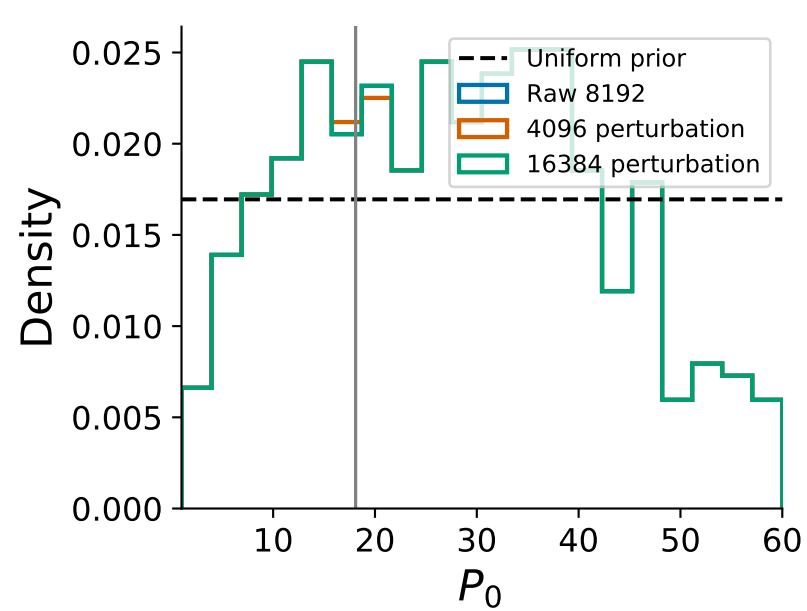


# Mstar-1sigma

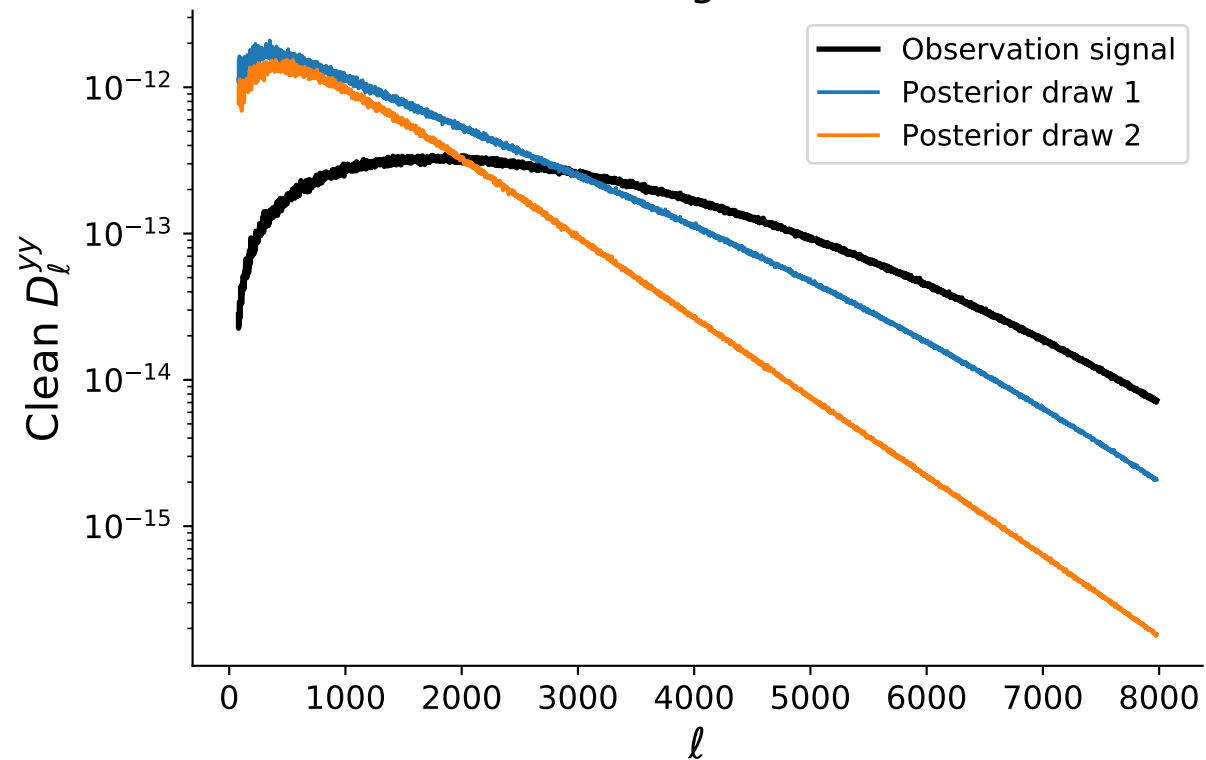


## fgas-8sigma

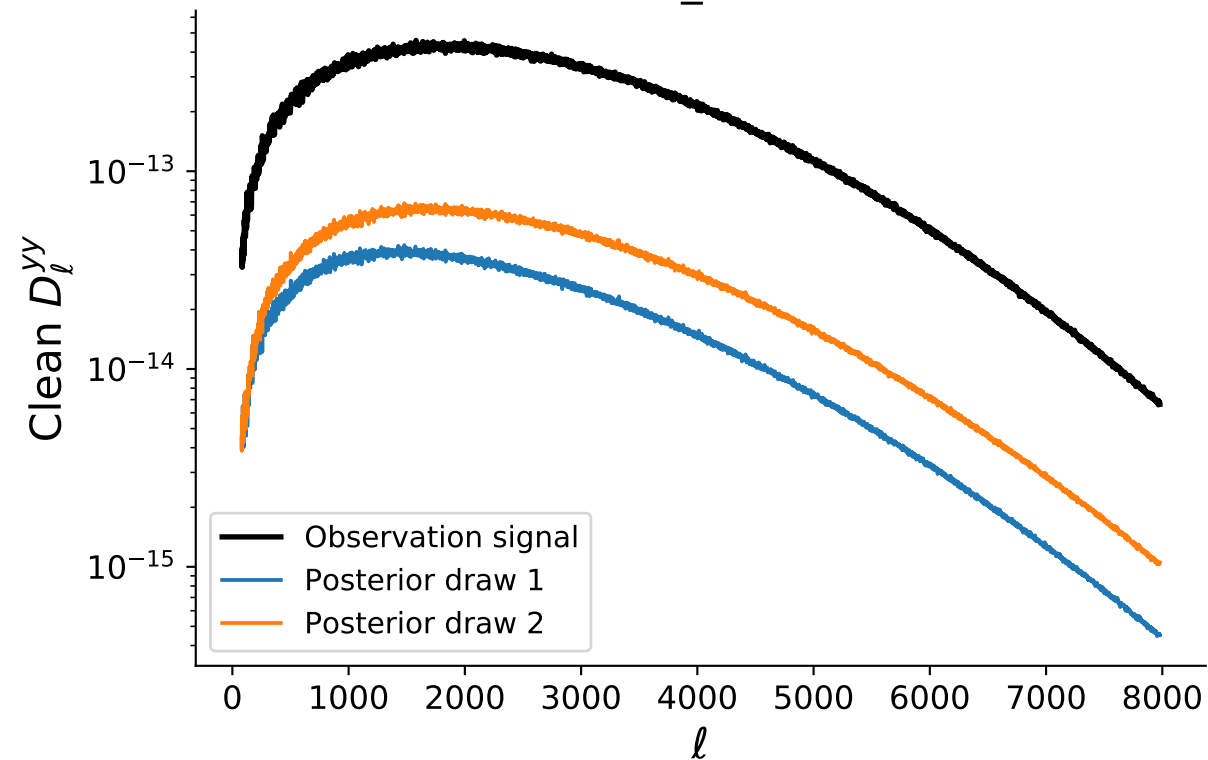




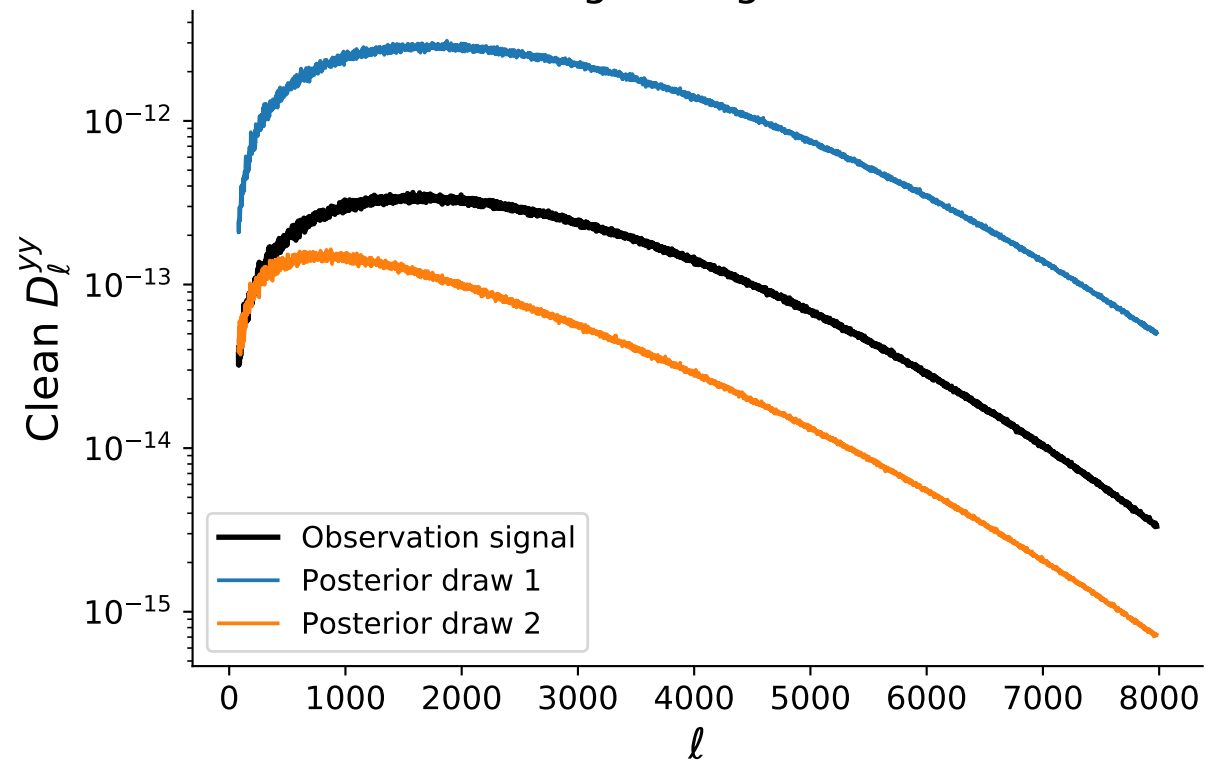
Battaglia12



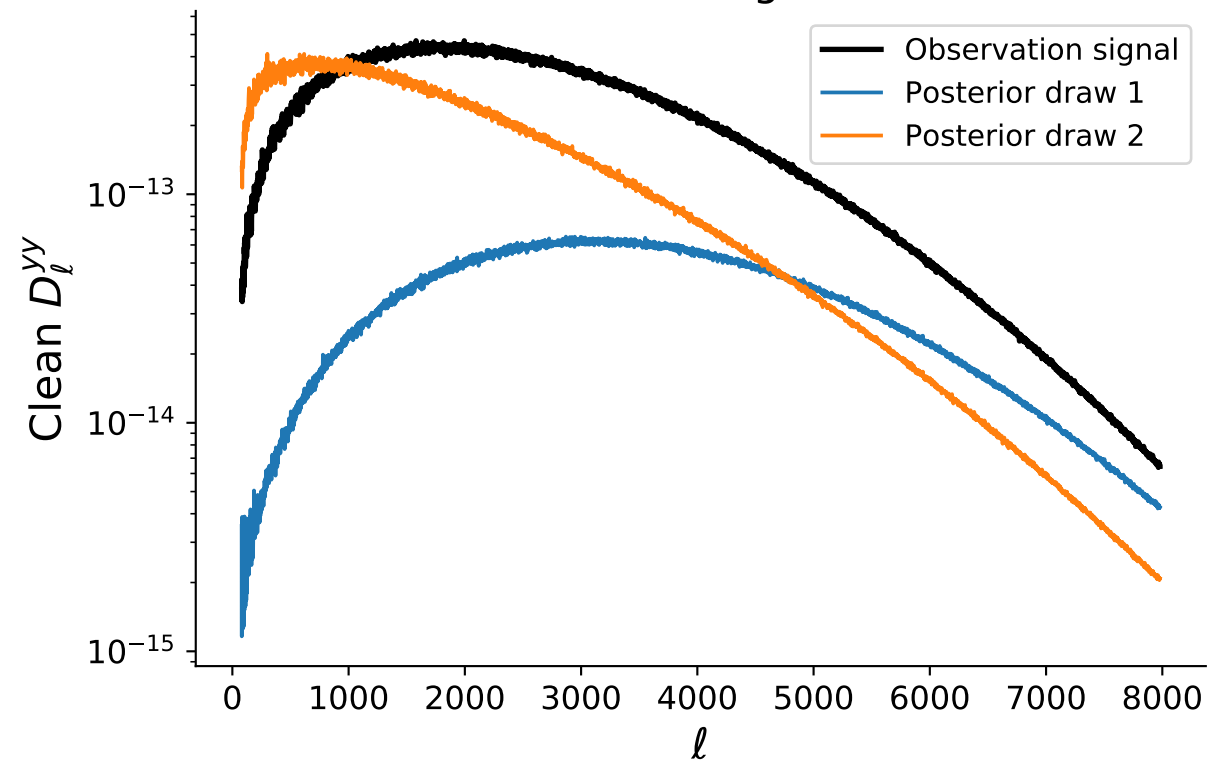
L1\_m9

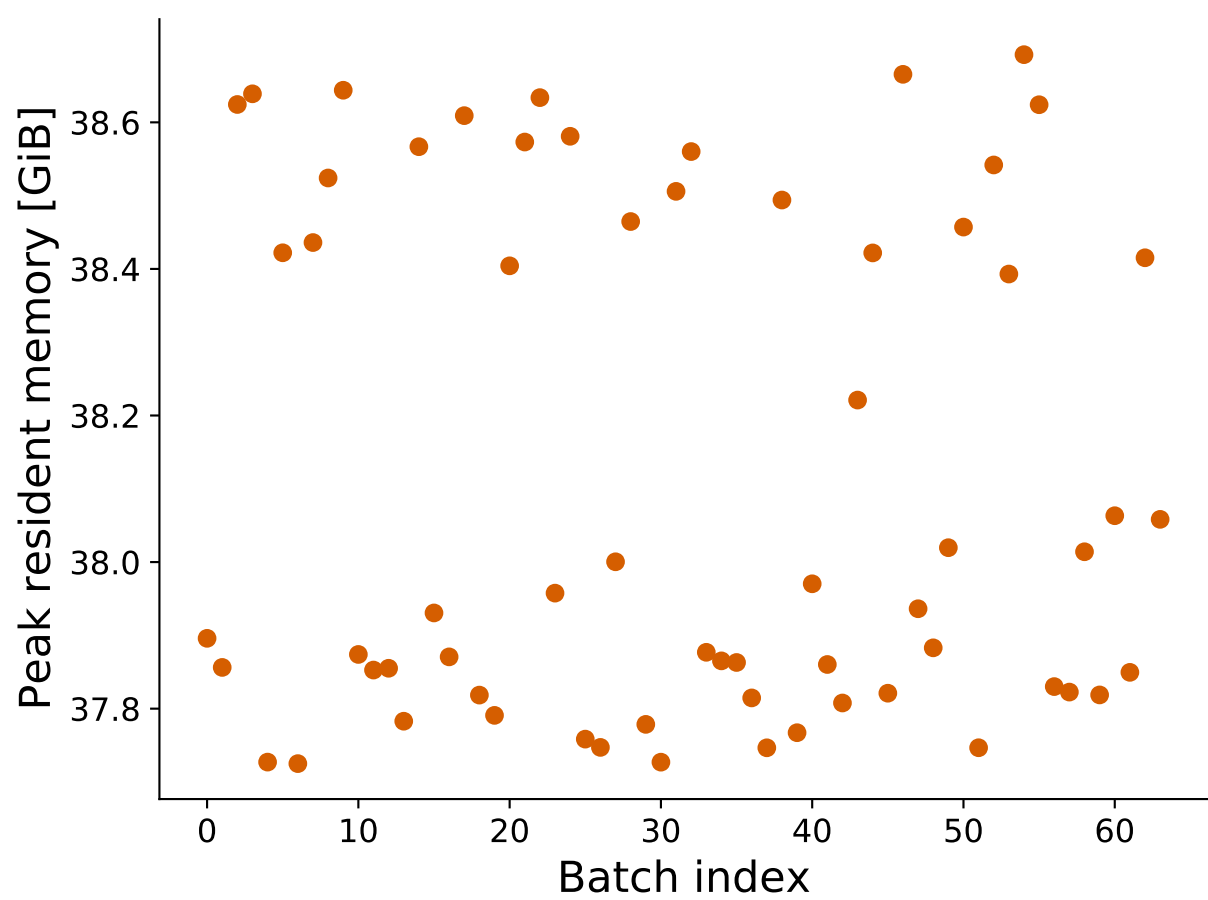
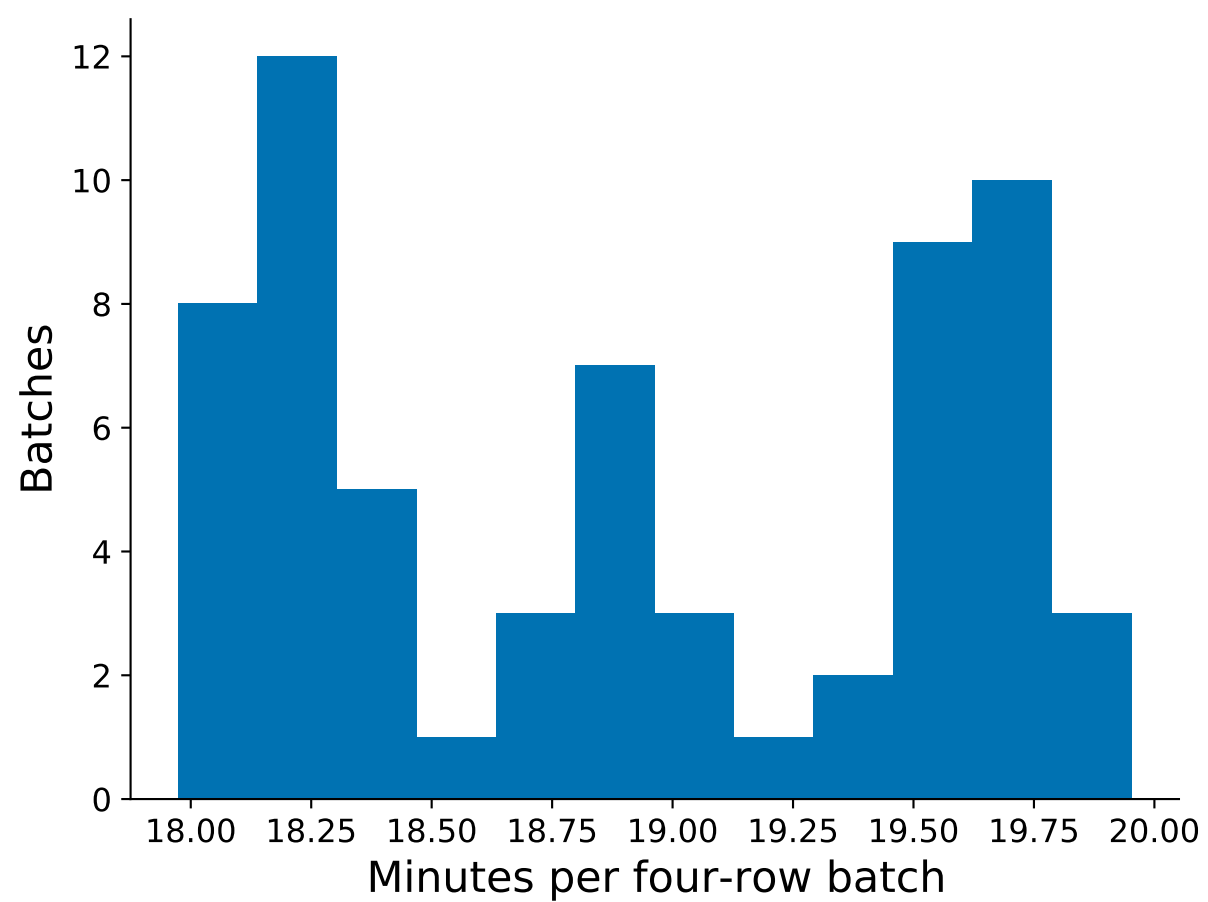


fgas-8sigma



Mstar-1sigma





# Interpretation and limits

Flat prior density does not guarantee informative data. A posterior resembling the prior can be correct for a weakly constrained parameter. It can also expose insufficient simulations, an unsuitable compression or a failed density estimator. The held-out RMSE, shuffled-data control, training-size trend and known B12 observation distinguish these possibilities better than posterior appearance alone.

The primary compressor multiplies all 7,900 raw  $D_{\text{ell}}$  values by the frozen nine-direction B12 finite-difference MOPED matrix. A signed asinh and standardization are then fitted using only optimization rows. A separate regression-based unbinned compressor transforms individual multipoles first and estimates derivatives from nearby training rows. It is an approximate alternative, not a global sufficiency certificate.

FLAMINGO is an external, out-of-family observation. Its posterior represents effective gNFW parameters for this fixed HalfDome painter, cosmology, radial truncation and estimator. It is not a measurement of a uniquely defined FLAMINGO gNFW truth. Observations use the saved fixed noise; training rows use independent noise realizations.

The coverage band is a binomial sampling reference for 64 marginal events. A Sobol design and correlations between parameters mean this is not an exact independent-binomial test of simulation-based calibration. A precise coverage study, field-to-field catalogue variance, foregrounds and a full survey covariance require additional work.

Smooth boundary means smooth continuation of the cached chord-mean pressure across the spherical edge. The physical gas still stops exactly at 4 R200c. Halving the cache changes interpolation node density, not the physical model. Greedy scheduling changes which thread paints a halo block, while shared geometry reuses positions, radii and intersected pixels across four different pressure fields.

The noise split convention is two independent maps each with the specified SO  $N_{\text{ell}}$ . If that curve describes a full-depth survey, two half-exposure observational splits ordinarily need twice that  $N_{\text{ell}}$  each. The current convention is retained and explicitly reported; this diagnostic is not a validation of survey split-depth modelling.